Analysis of auto-fluorescence in NHP baseline in vivo retinal images

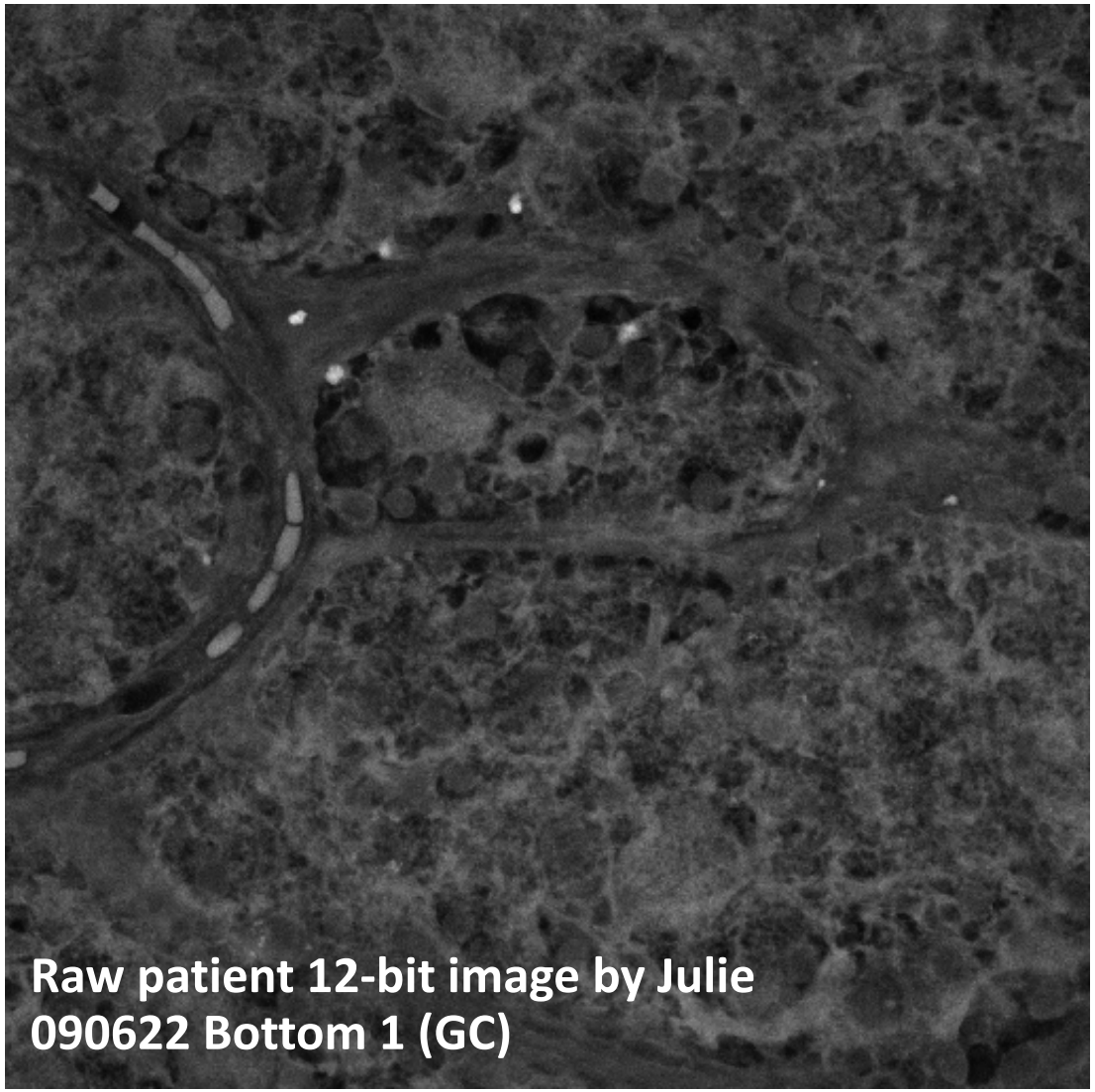
Alex Matov
November 2022

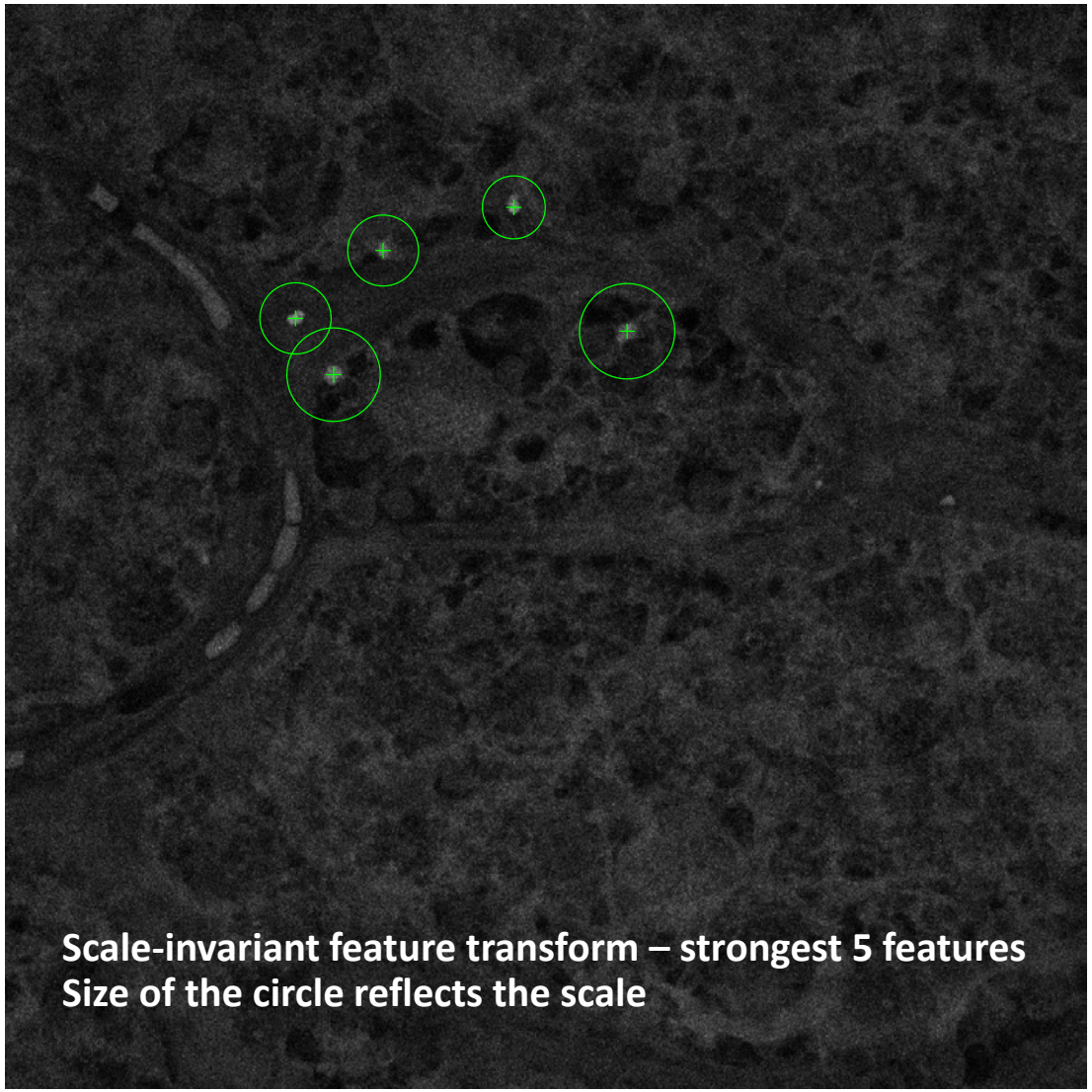
Summary of data & findings

- 13 NHP in vivo datasets/eye samples
- 6-8 images time-lapse images per sample, split roughly half prior to & post injection
- 15 faint auto-fluorescent blobs were detected, an average of ~1.15 per sample (max. 3 in an eye)
- Aggregates had higher contrast & were more easily detected in images acquired prior to injection
- The blobs are often circular with peak intensity ~40[A.U.], perimeter ~200um, & area ~.003mm2
- Images acquired with 8-bit camera; intensity range is 0 to 255 [A.U.]; 1 image pixel = 7um in sample

Cyno 170395 \Cyno1_000.tif - 1
Cyno 180424 \Cyno1_004.tif - 0
Cyno 191797 \Cyno1_004.tif - 2
Cyno 191800 \Cyno1_004.tif - 2
Cyno 191815 \Cyno1_004.tif - 3
Cyno 191817 \Cyno1_004.tif - 1
Cyno 191823 \Cyno1_004.tif - 1
Cyno 200188 \Cyno 200188_004.tif - 0
Cyno 200188 \Cyno2_005.tif - 1
Cyno 200190 \Cyno2_003.tif - 0
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Cyno 200217 \Cyno2_003.tif - 2
Cyno 200251 \Cyno2_006.tif - 2

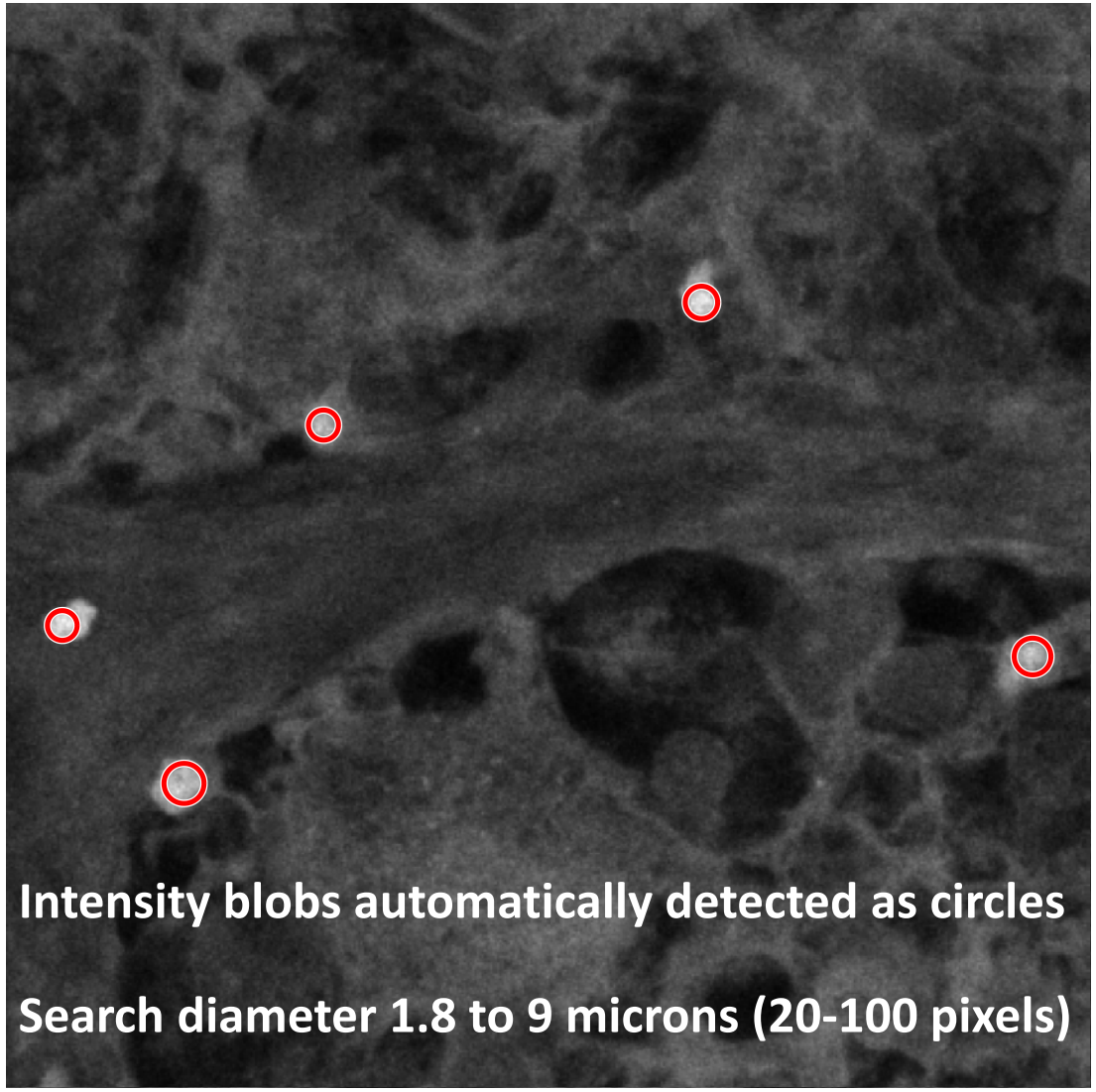
Example: Glaucoma SDEB Eye #2 – SIFT features test on ex vivo data

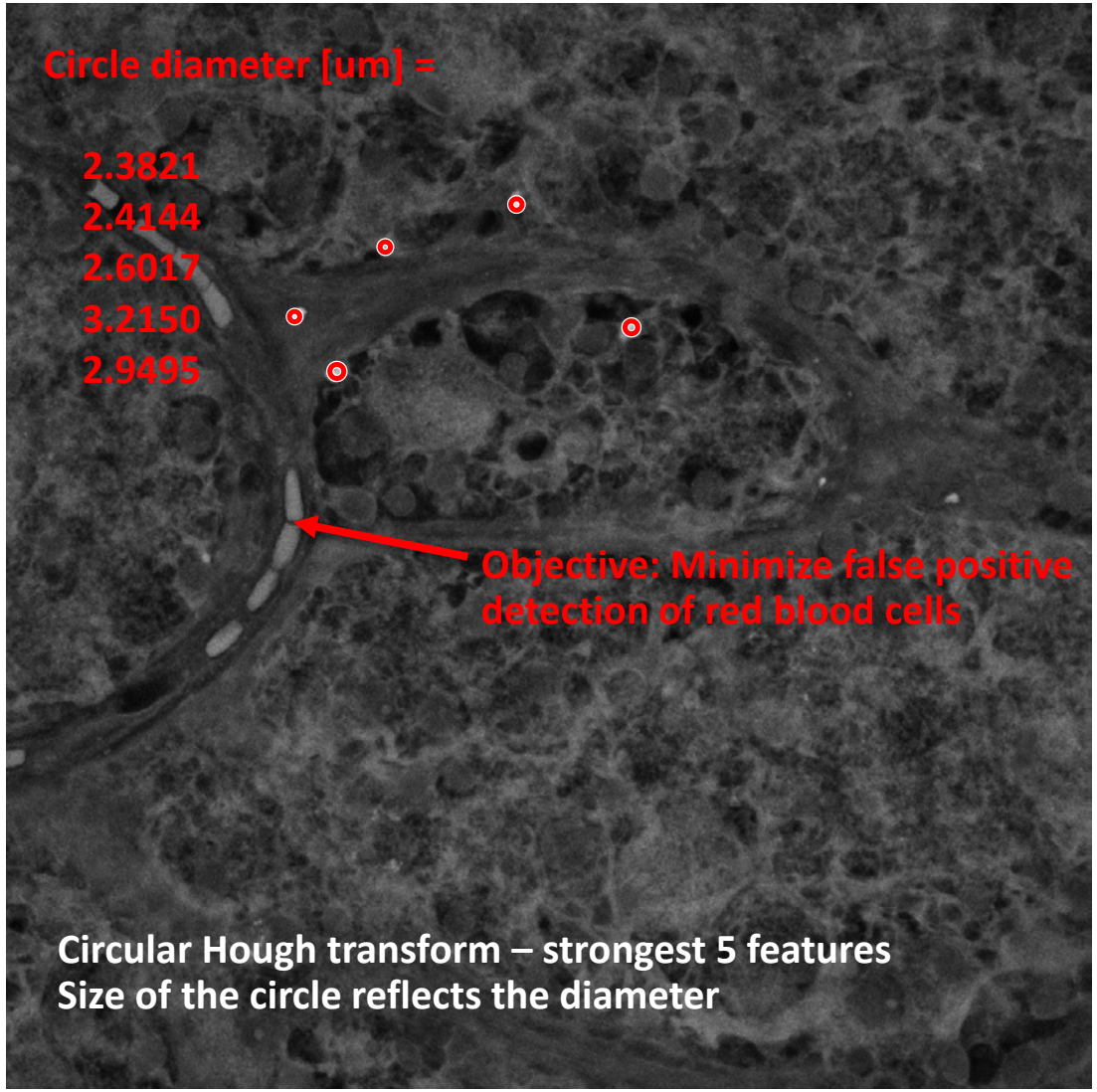




SIFT detects distinct features in ex vivo GC image data

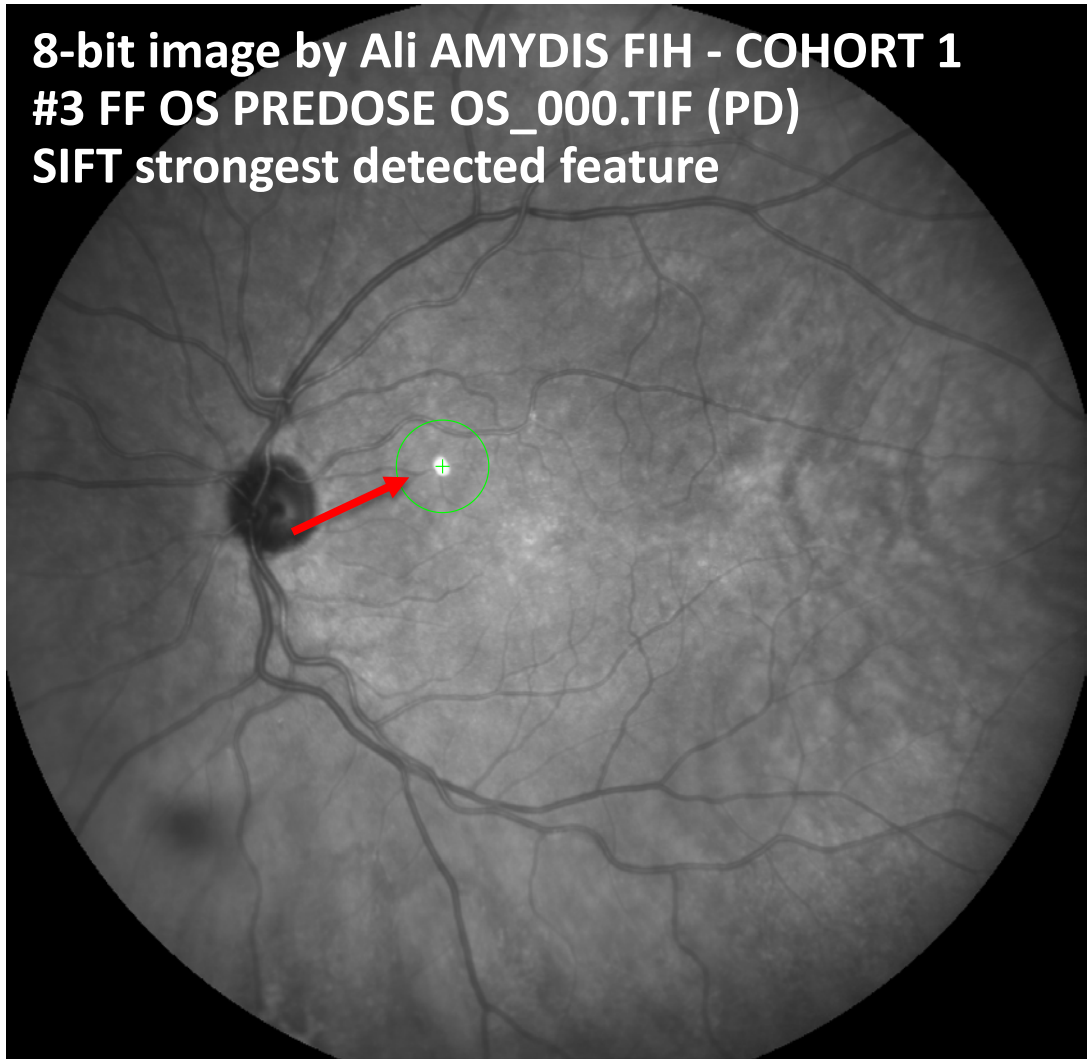
Example: Glaucoma SDEB Eye #2 – Circular Hough transform test on ex vivo data

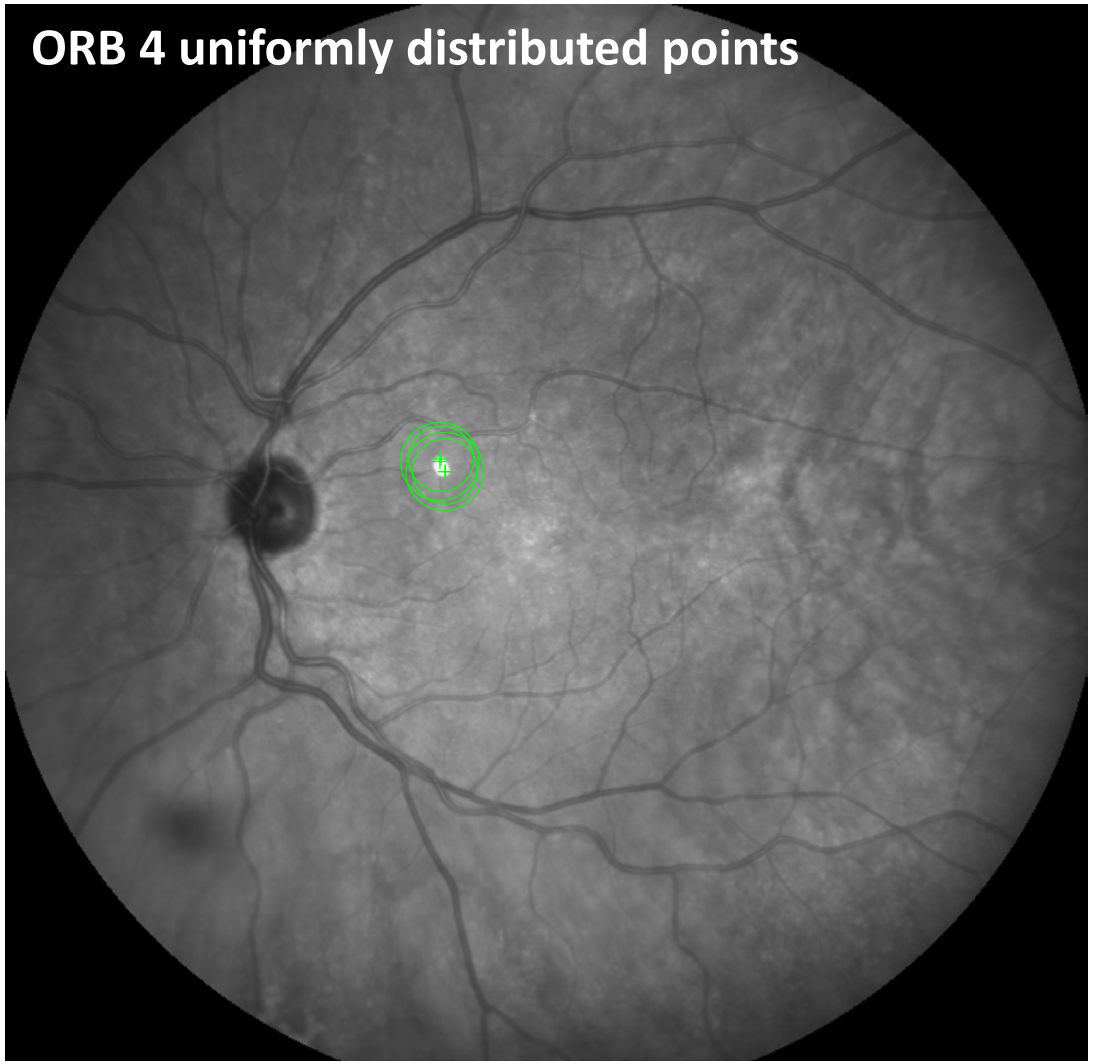




Circular Hough transform fits circles to round bright patches in ex vivo GC image data

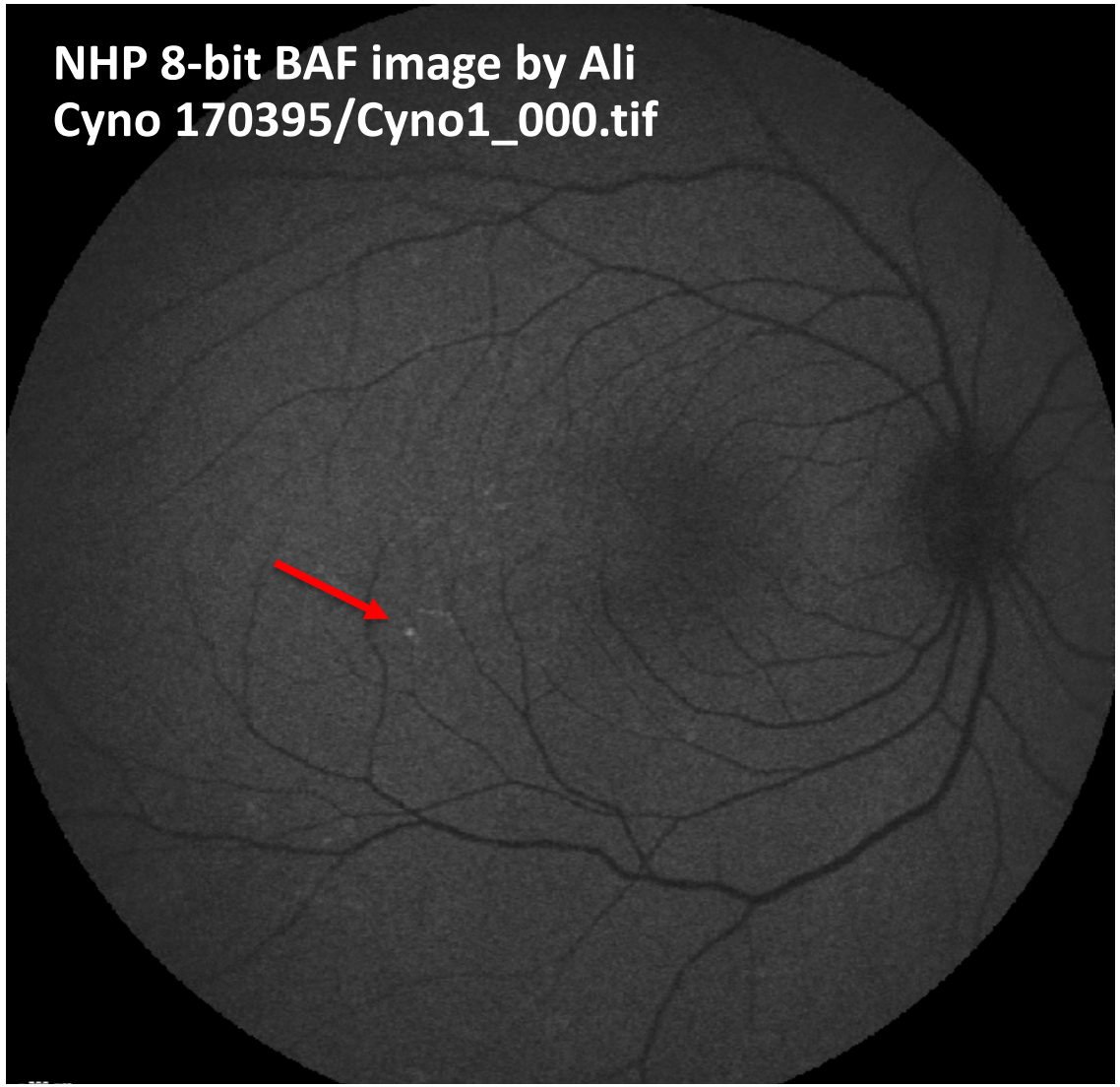
Pre-dose (no labeling) aggregates in in vivo human PD

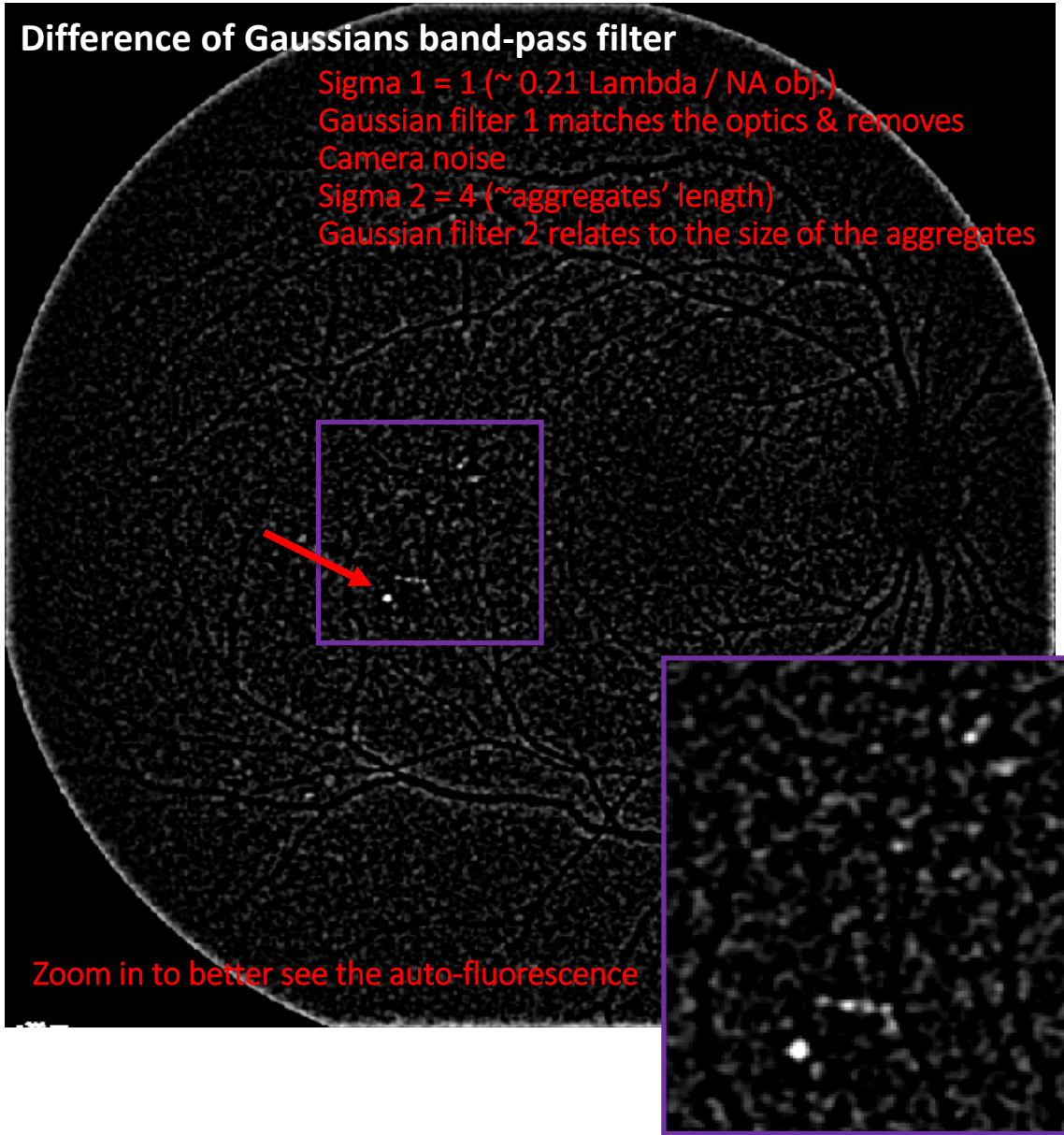




SIFT & ORB test on in vivo PD image data pre-injection – perhaps could be used in a classification test as positive samples

Preliminary analysis

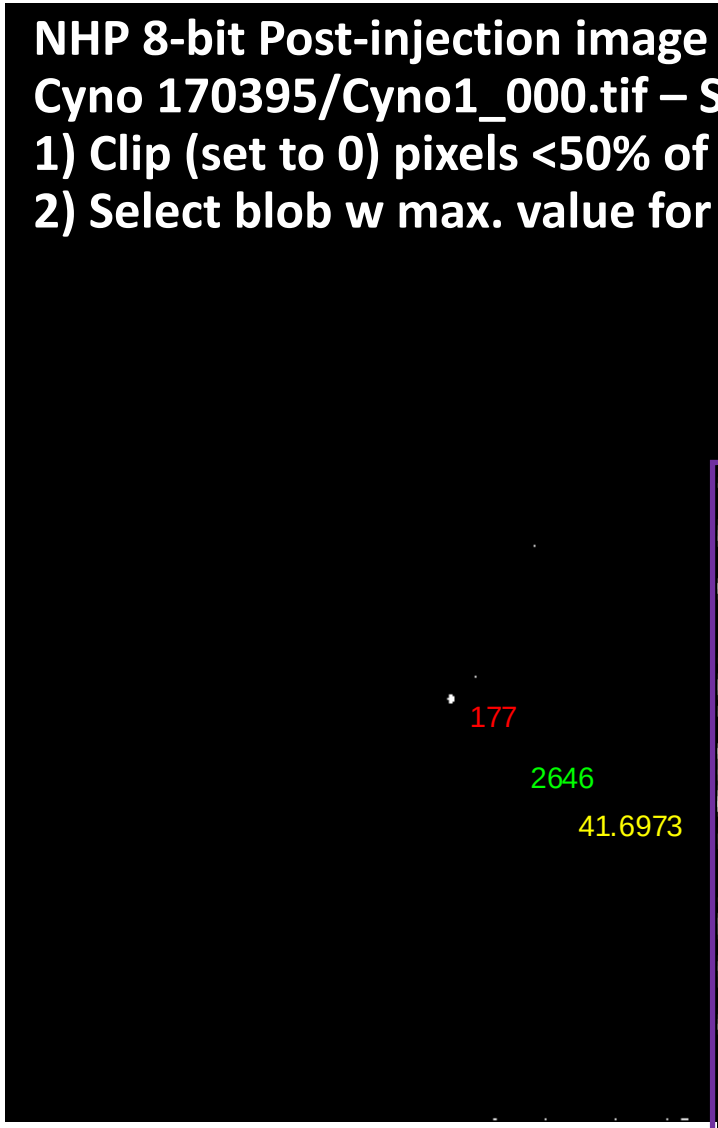


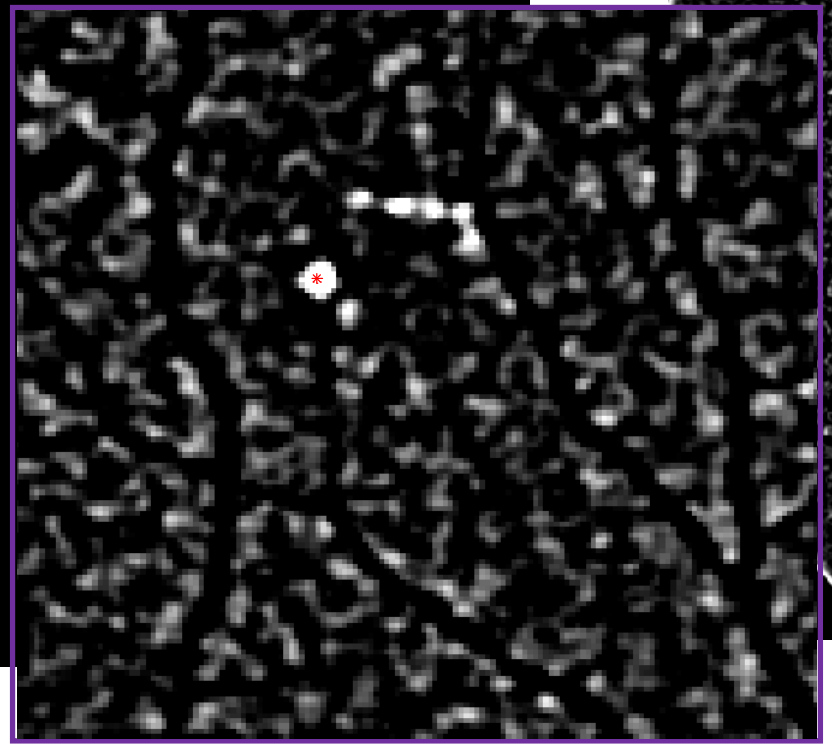


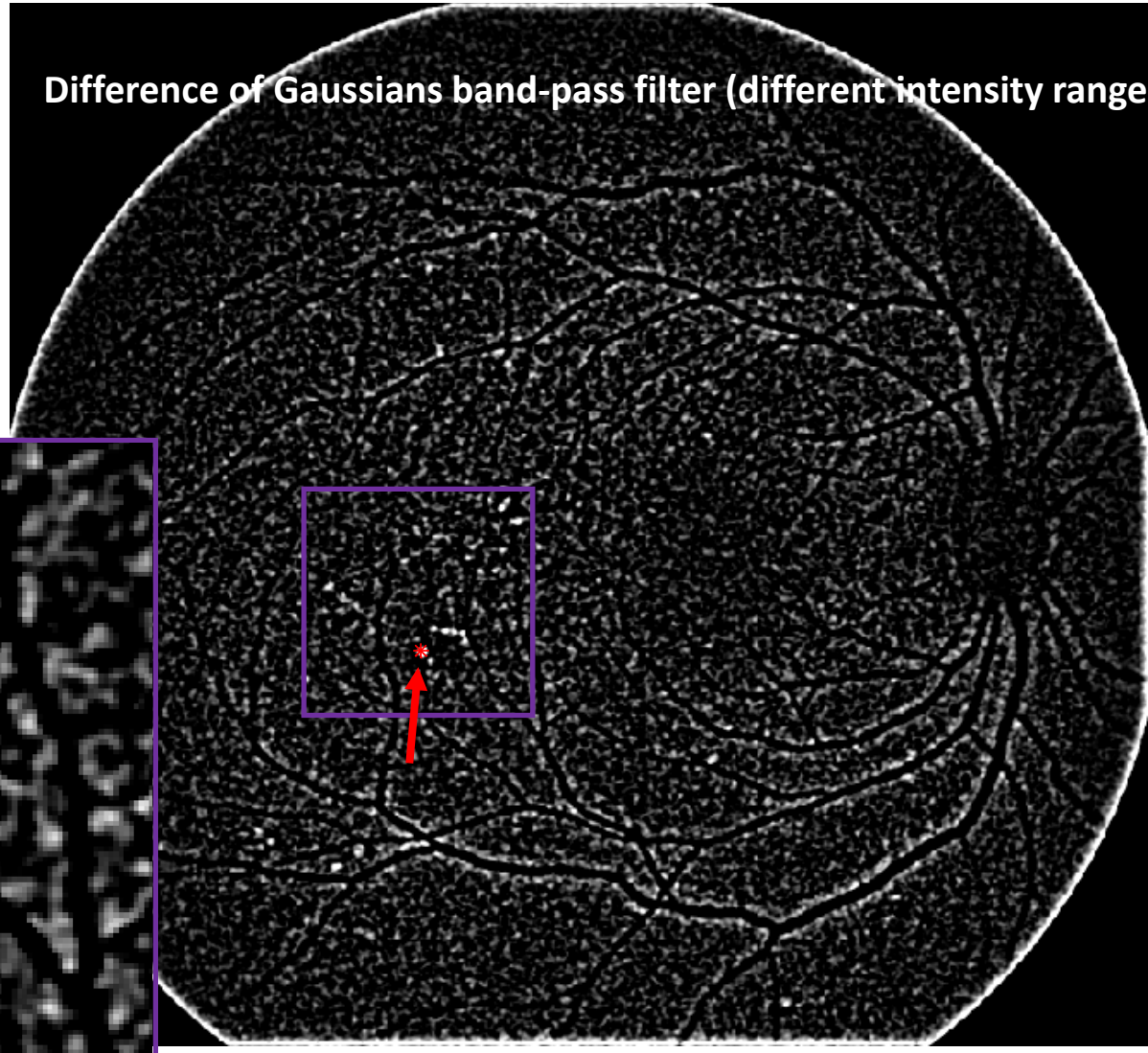
Preliminary analysis

perimeter in microns (red), area in square microns (green), max. intensity in arbitrary units (yellow)

NHP 8-bit Post-injection image by Ali
Cyno 170395/Cyno1_000.tif – Segmentation:
1) Clip (set to 0) pixels <50% of max. intensity of DoG
2) Select blob w max. value for peak intensity



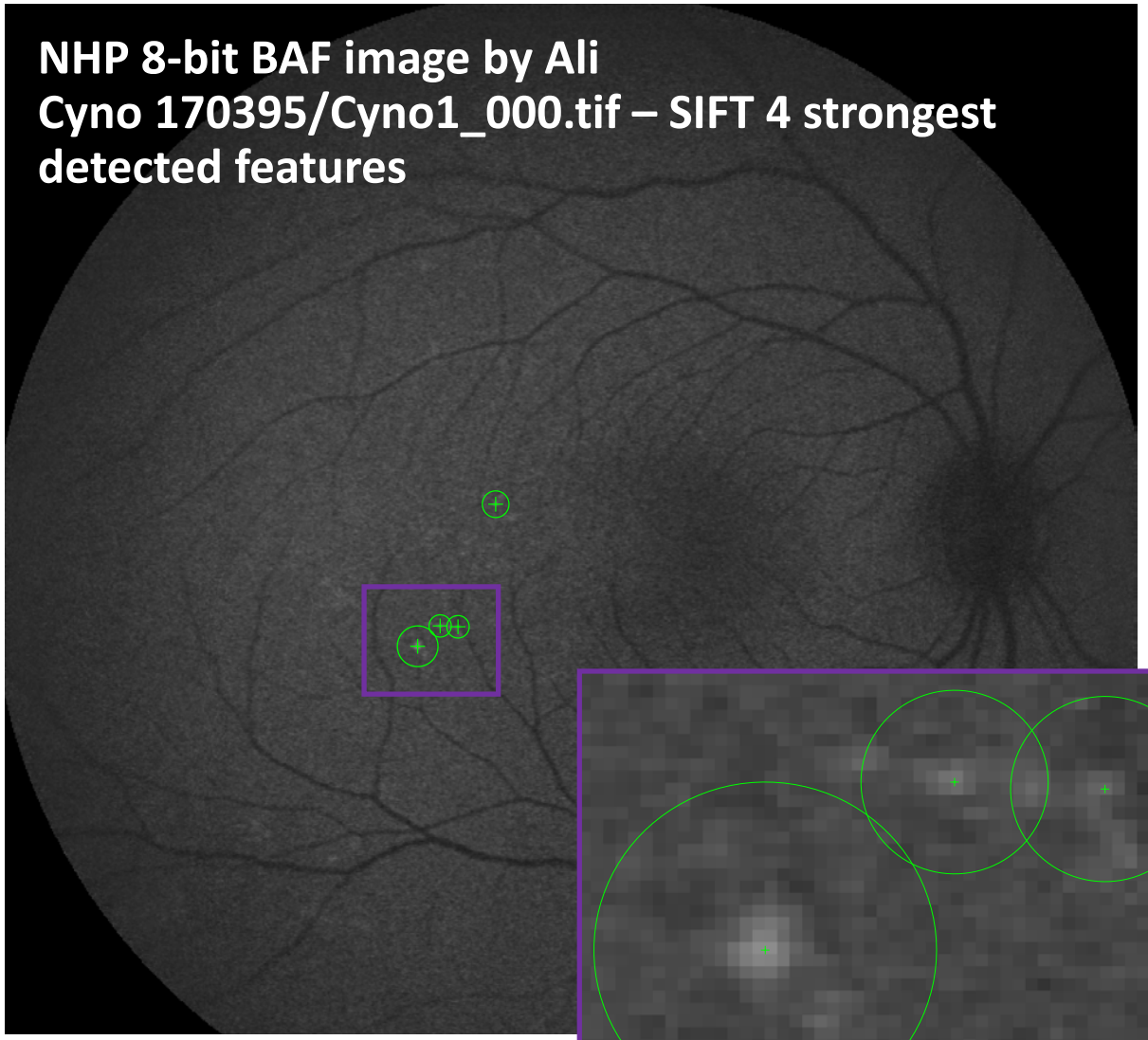


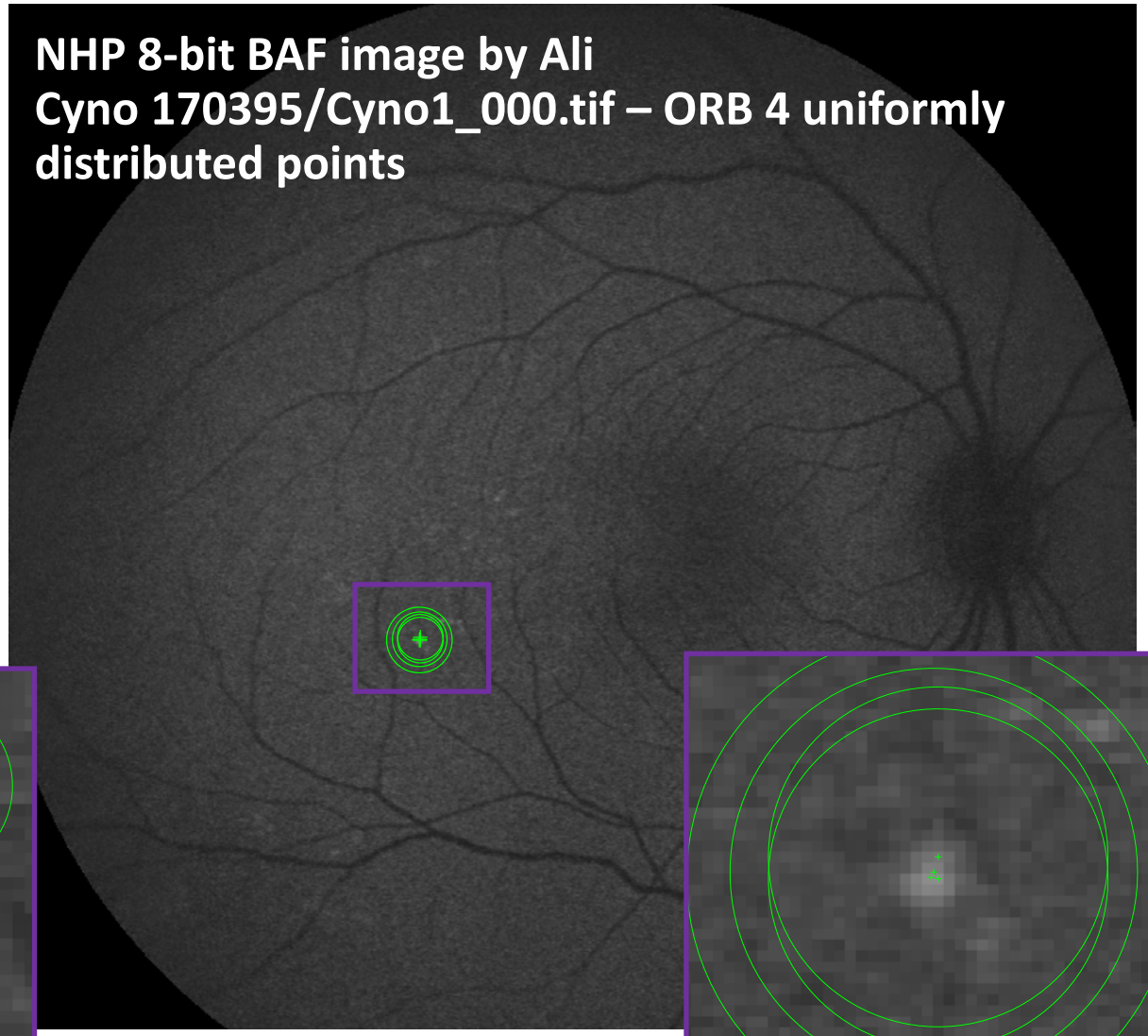


0.002646 square millimeters

Zoom in to better see the auto-fluorescence

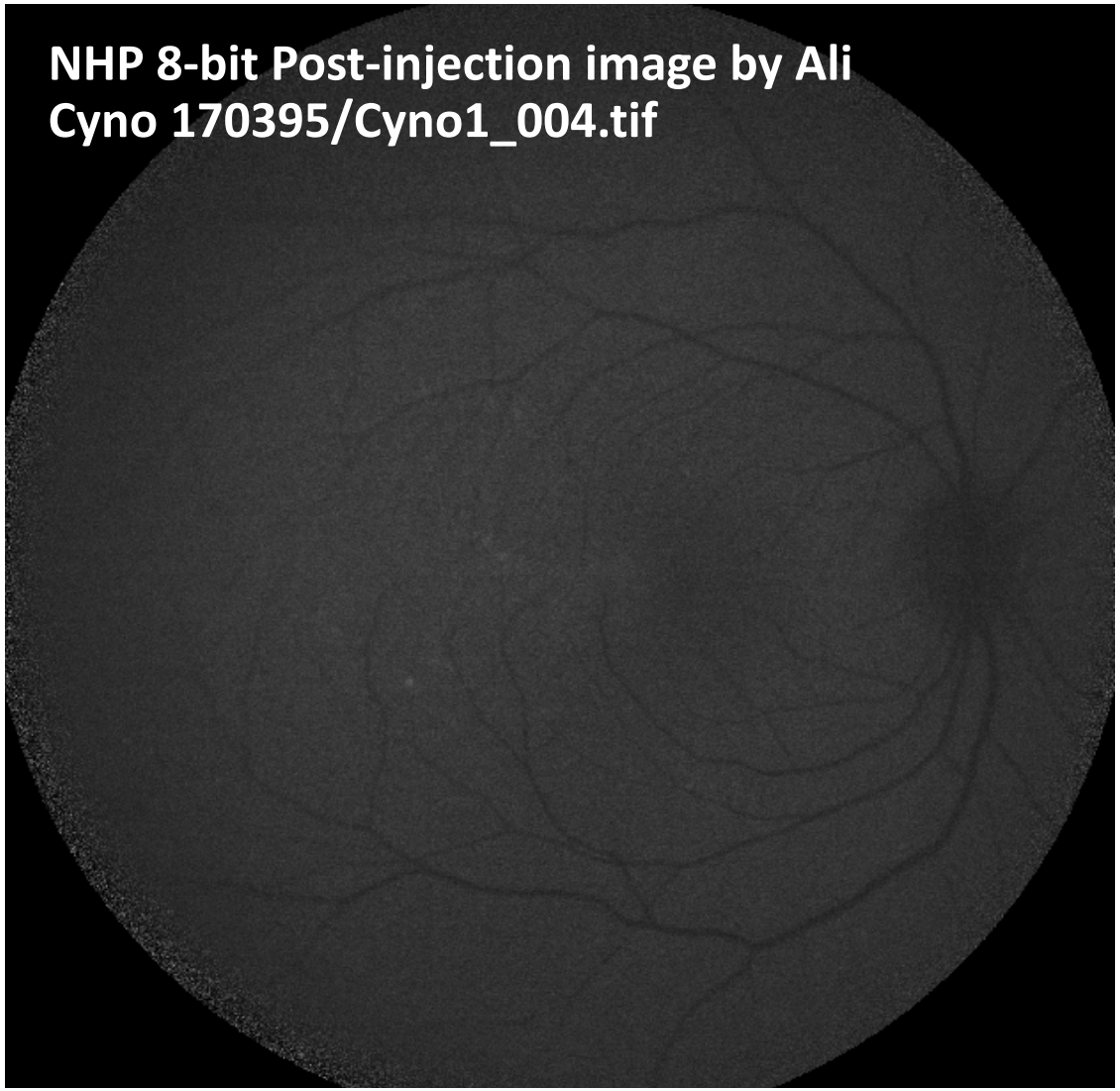
Preliminary analysis

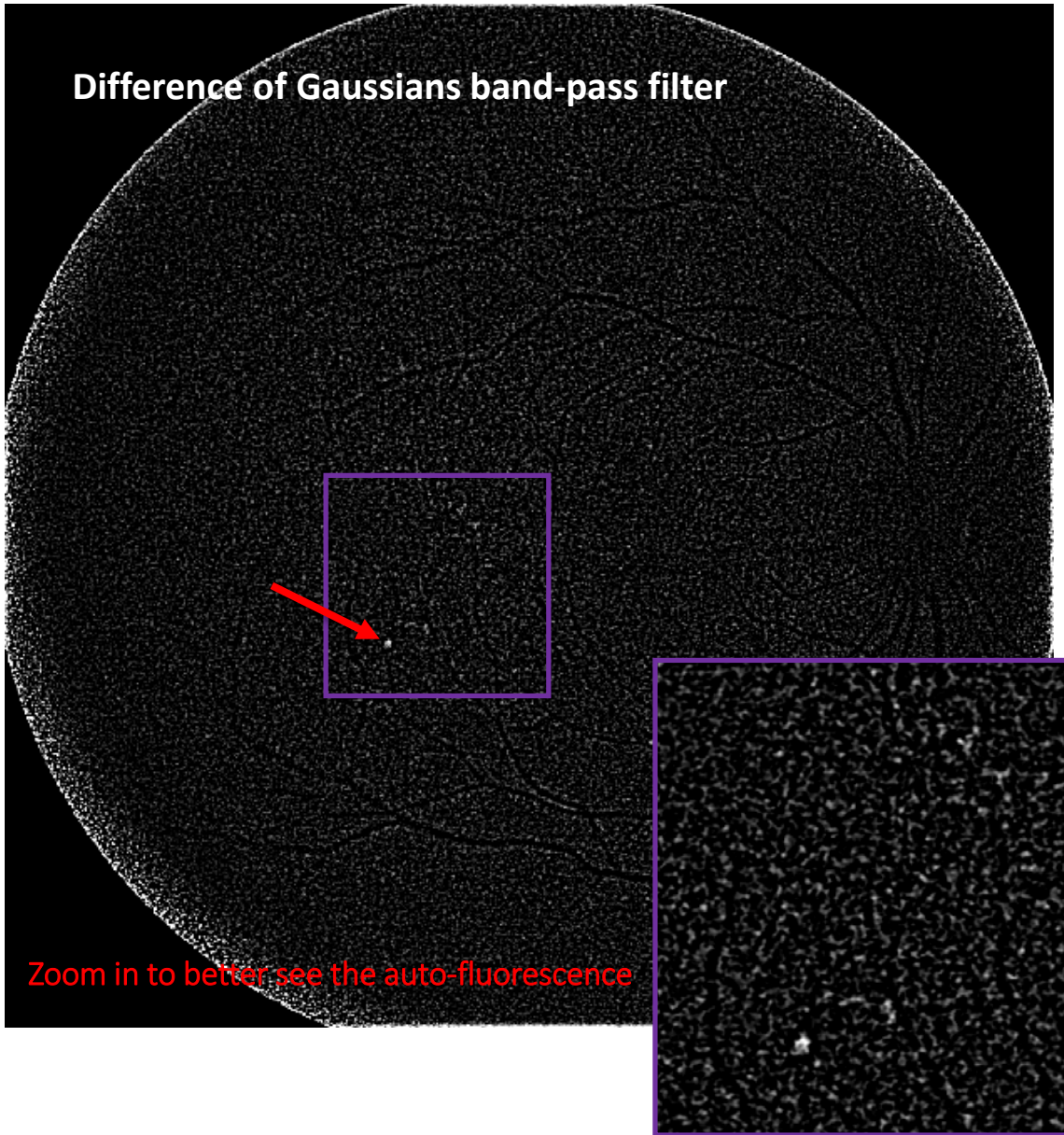




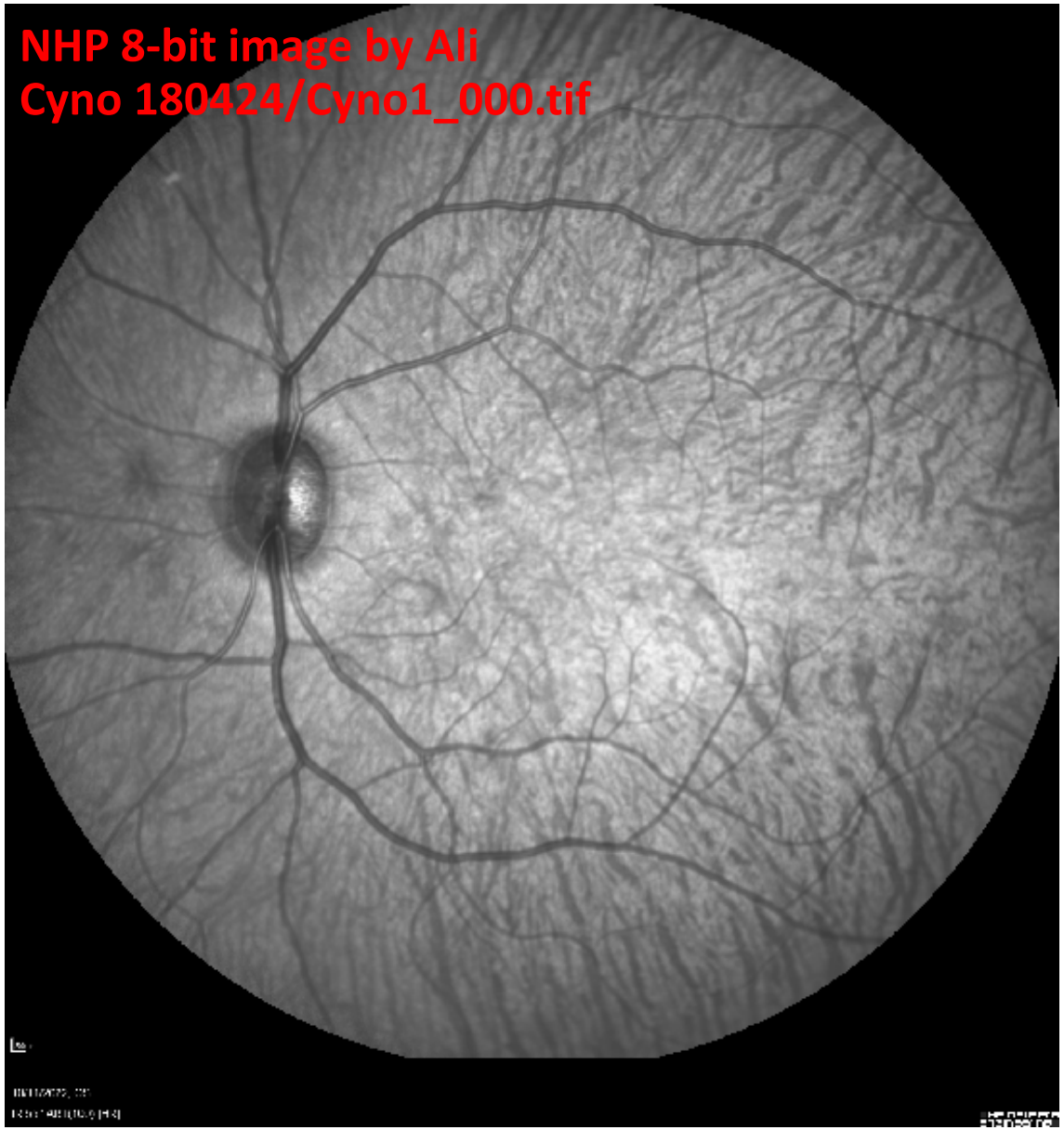
Zoom in to better see the SIFT detections

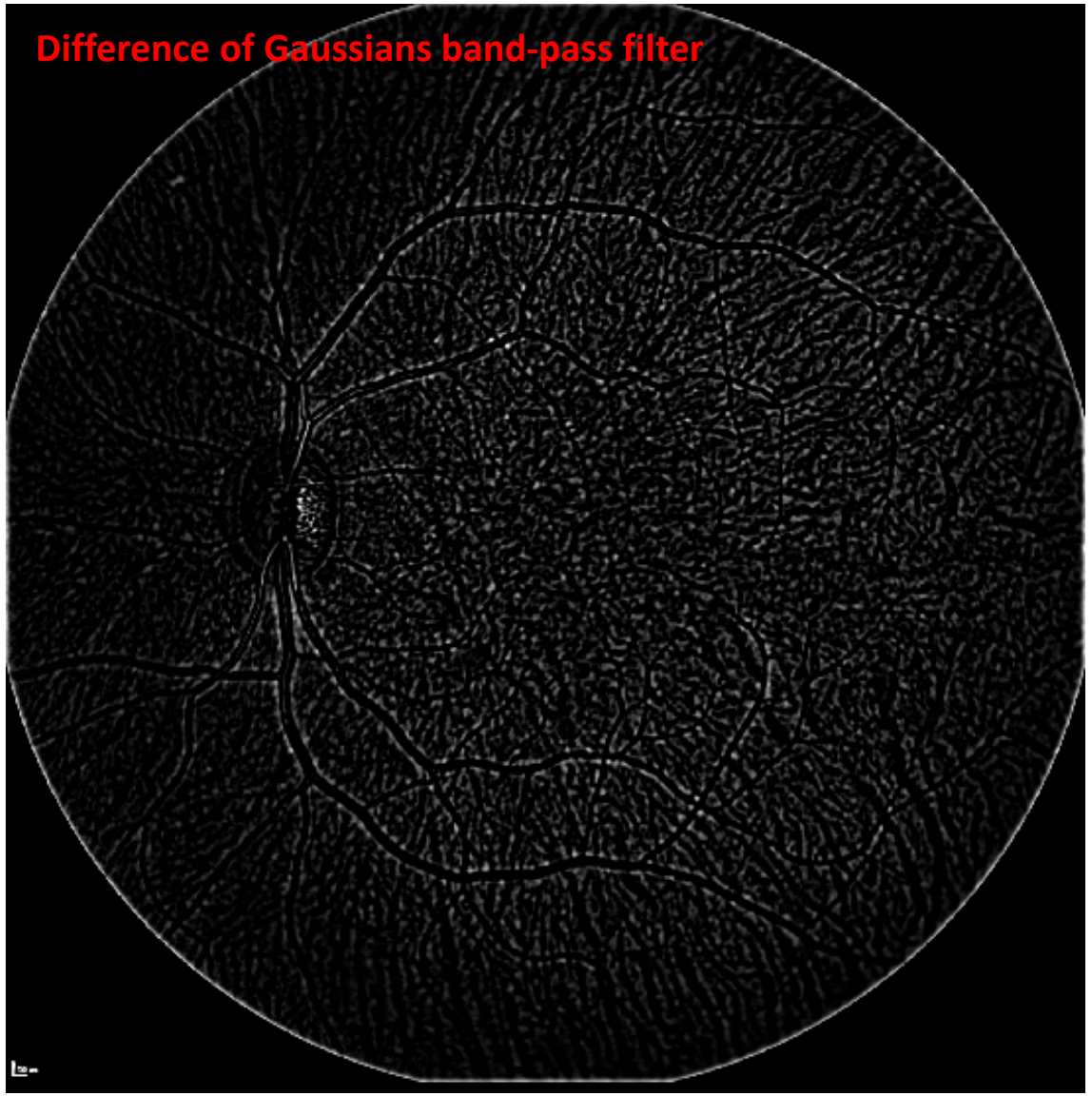
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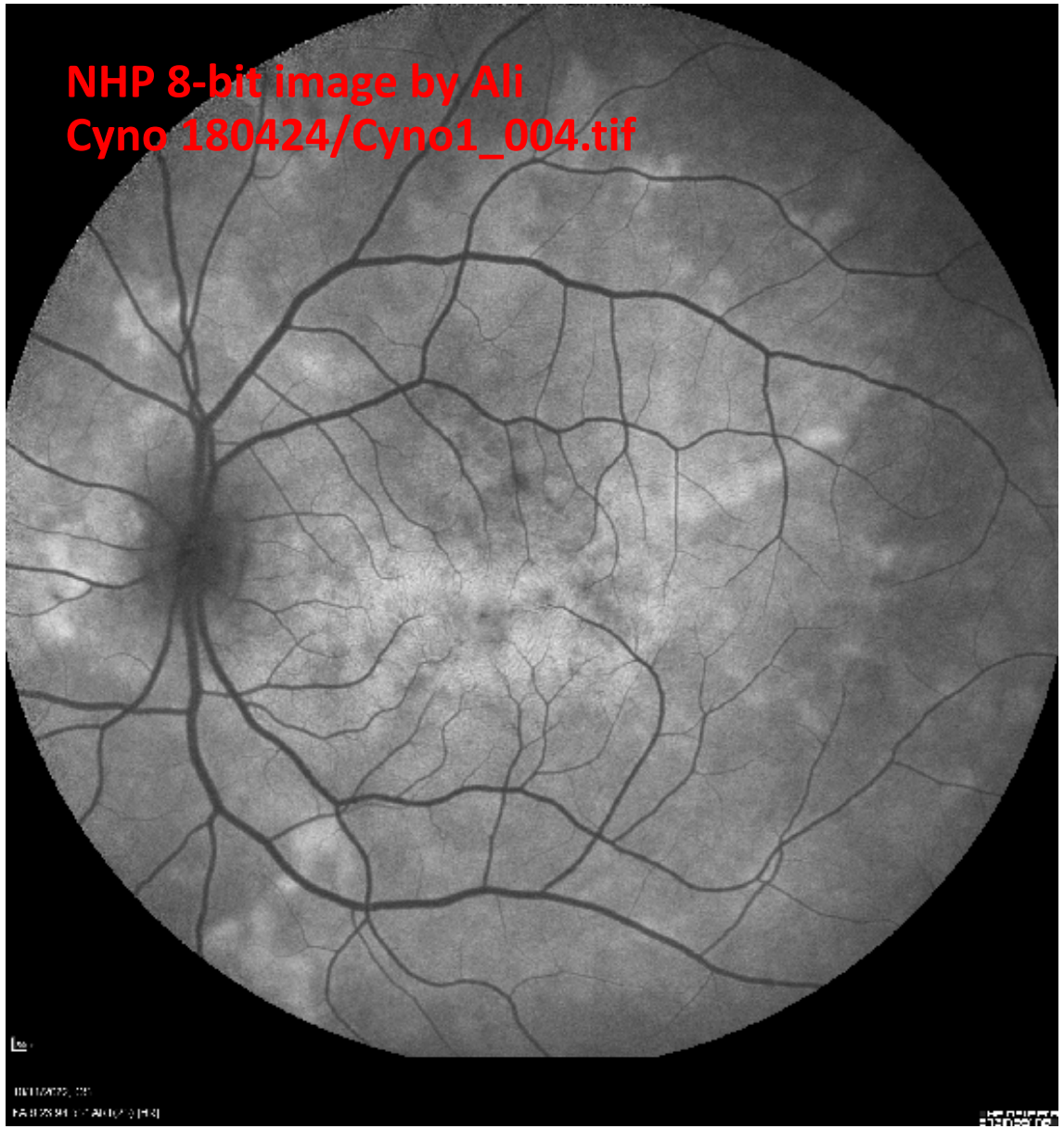


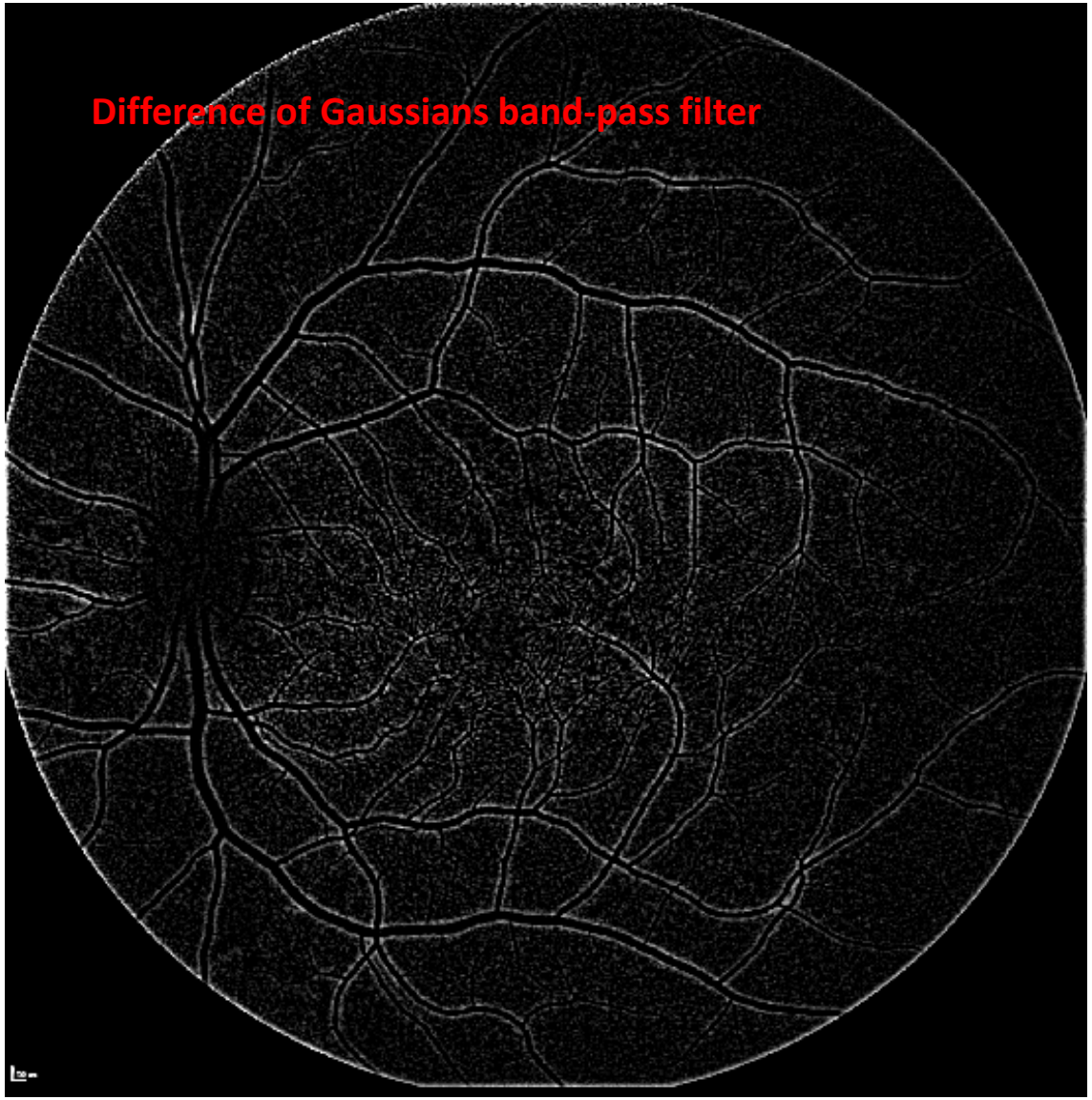
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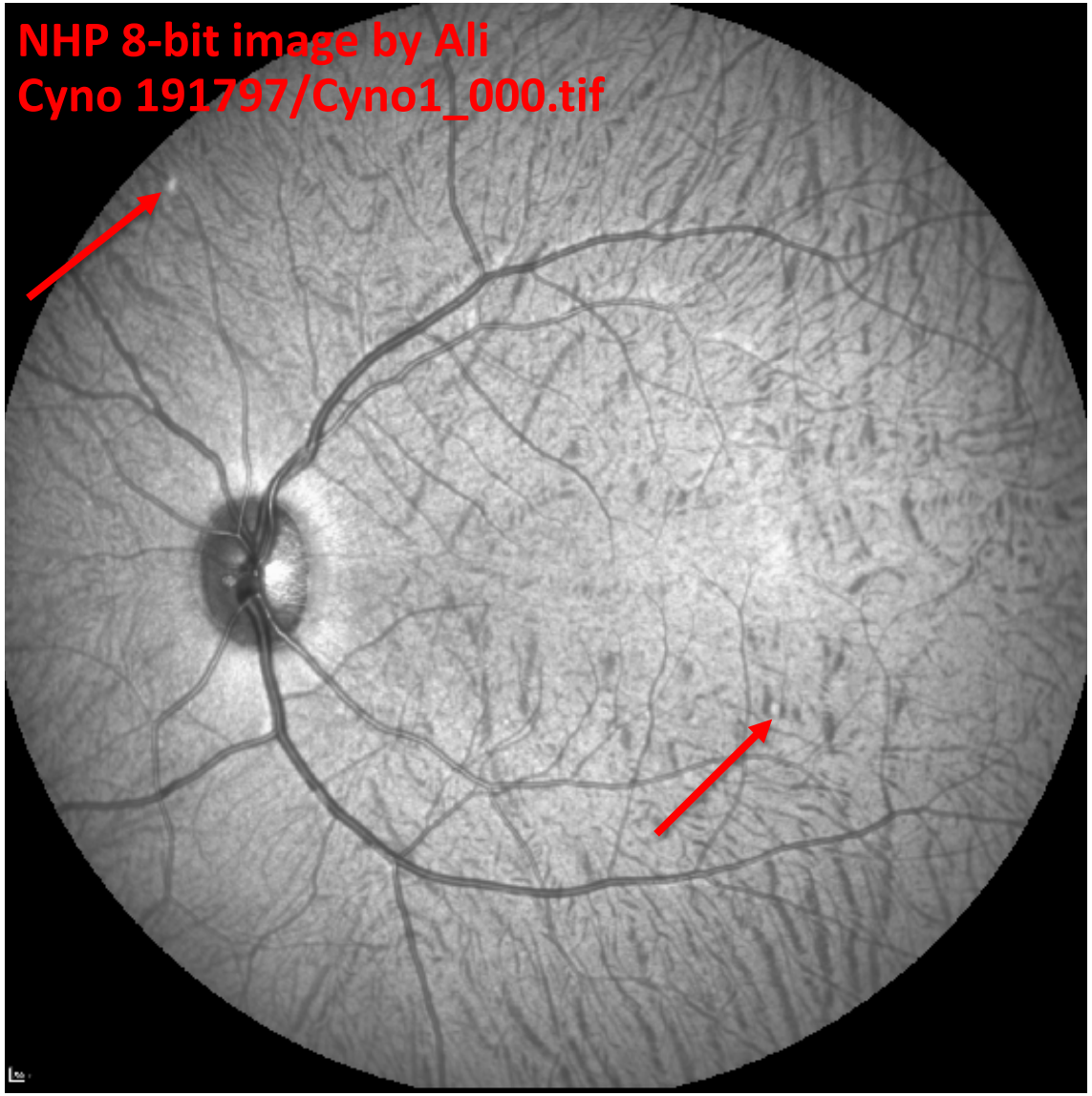


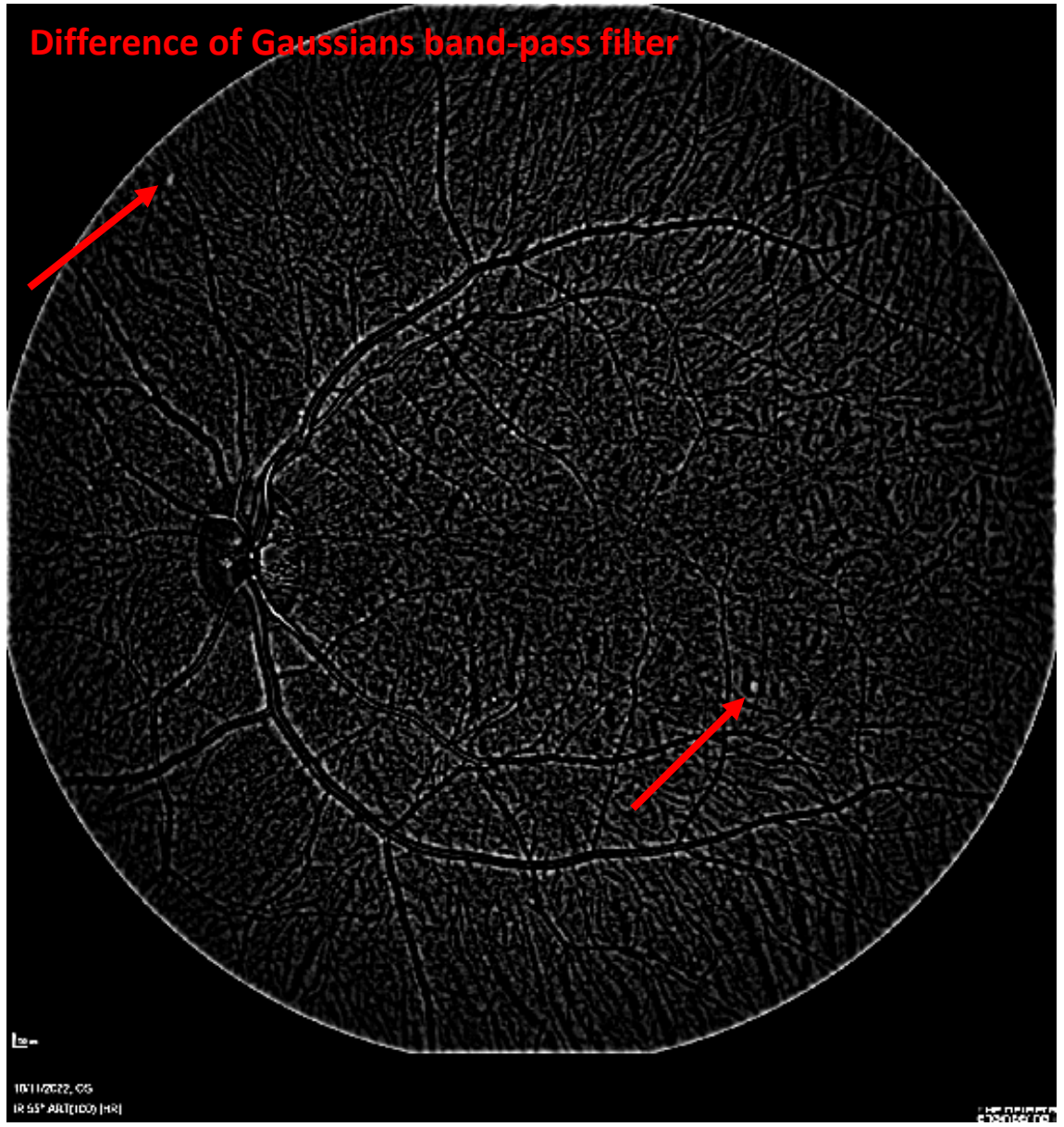
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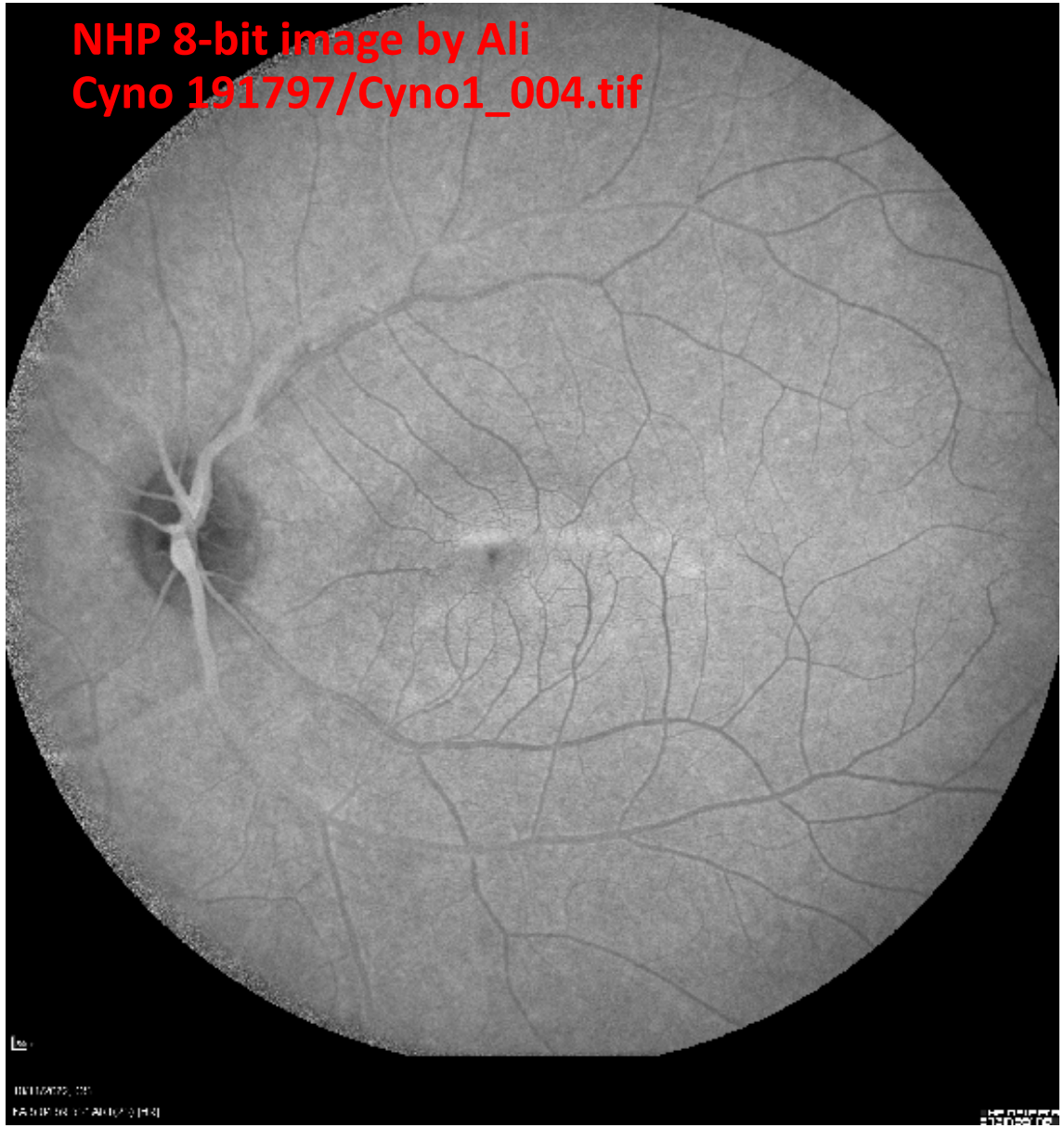


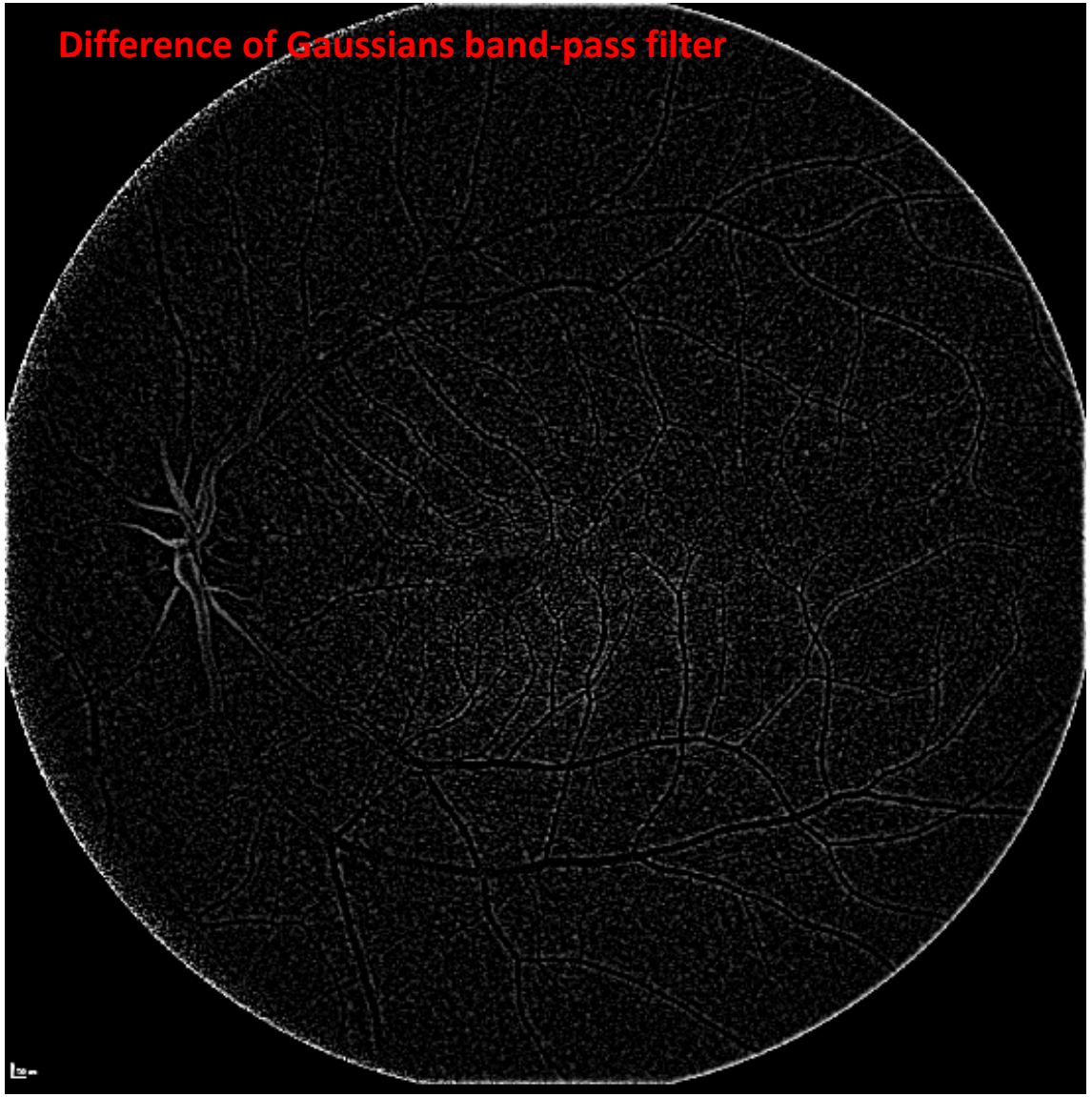
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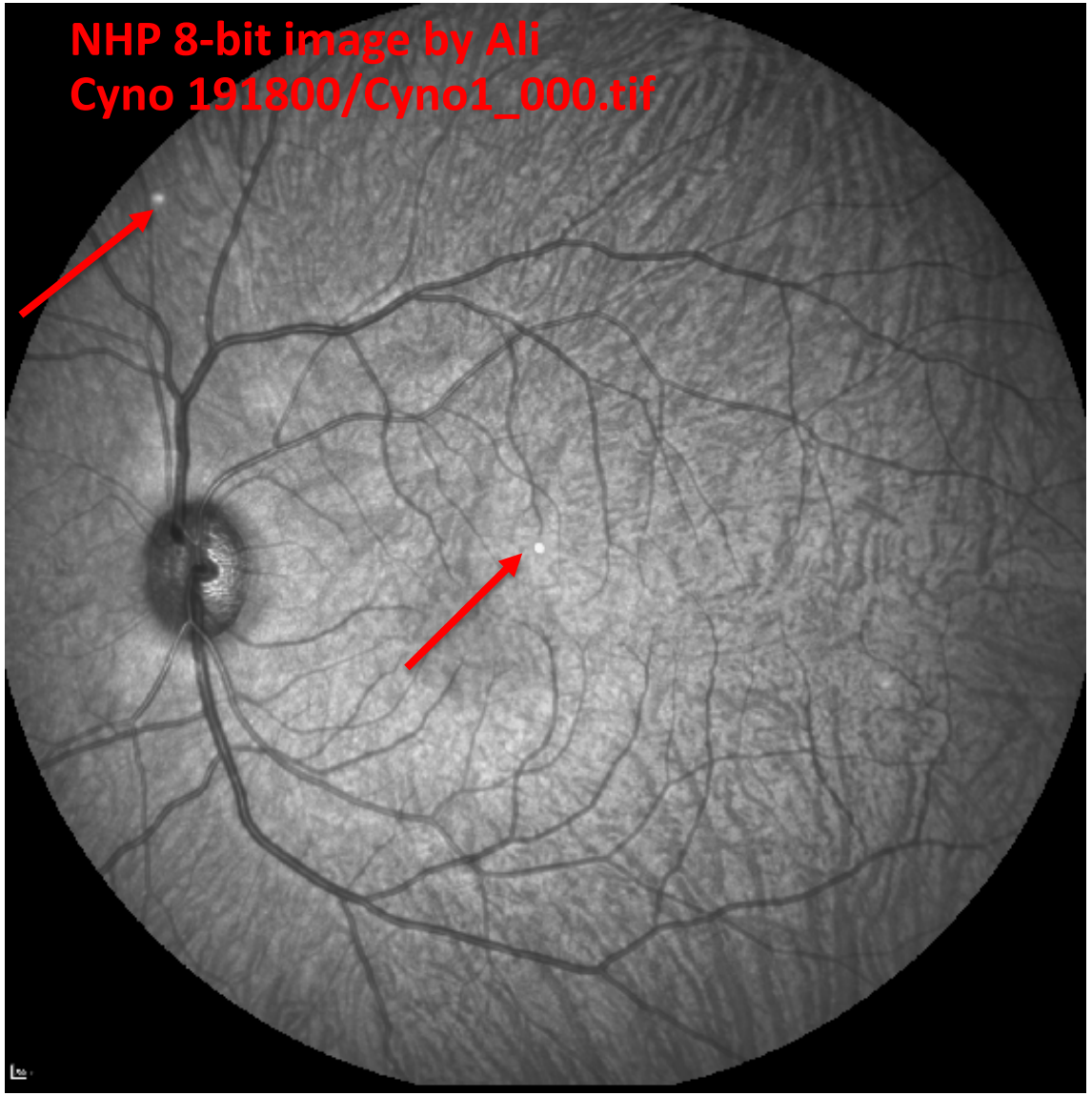


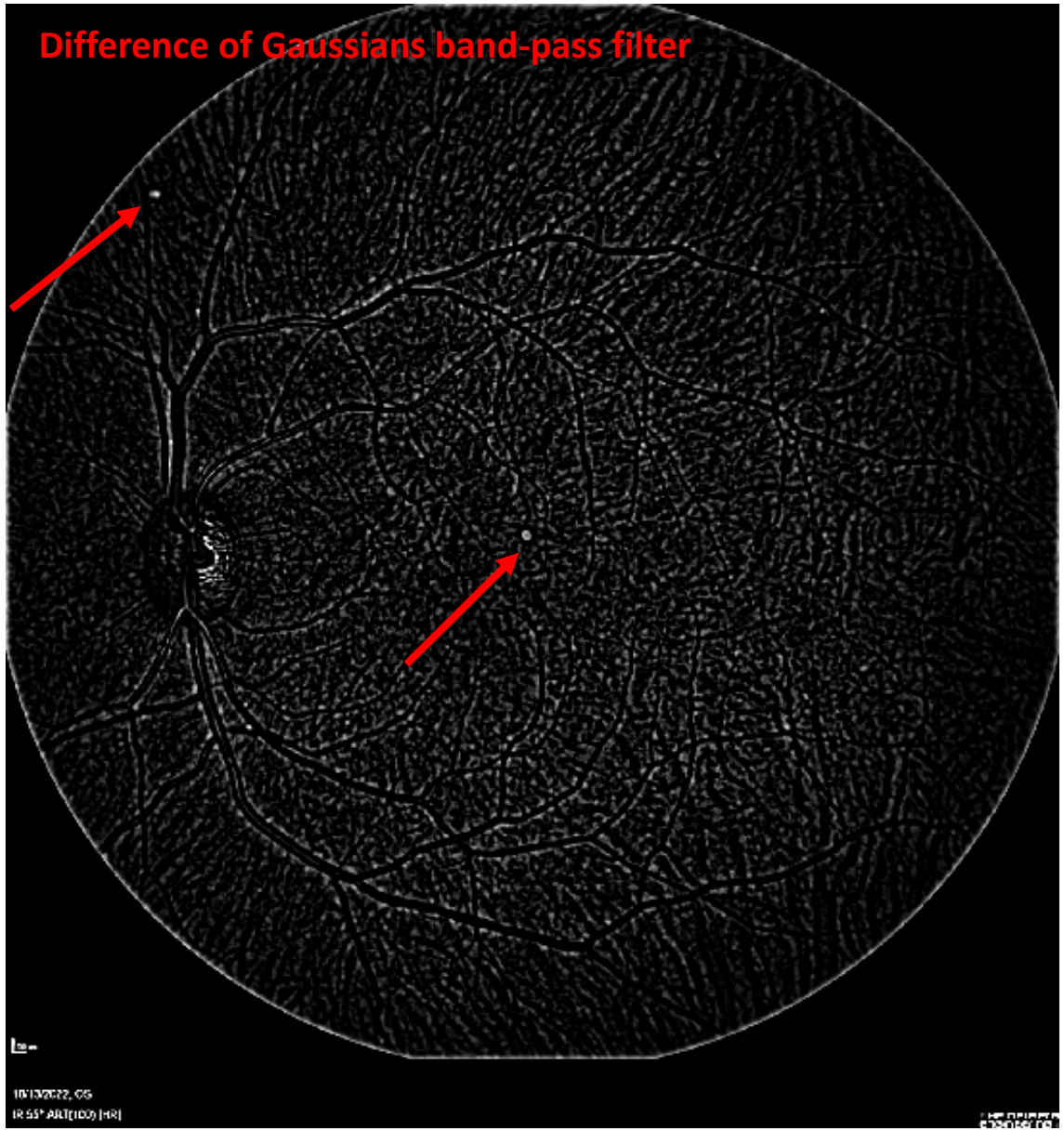
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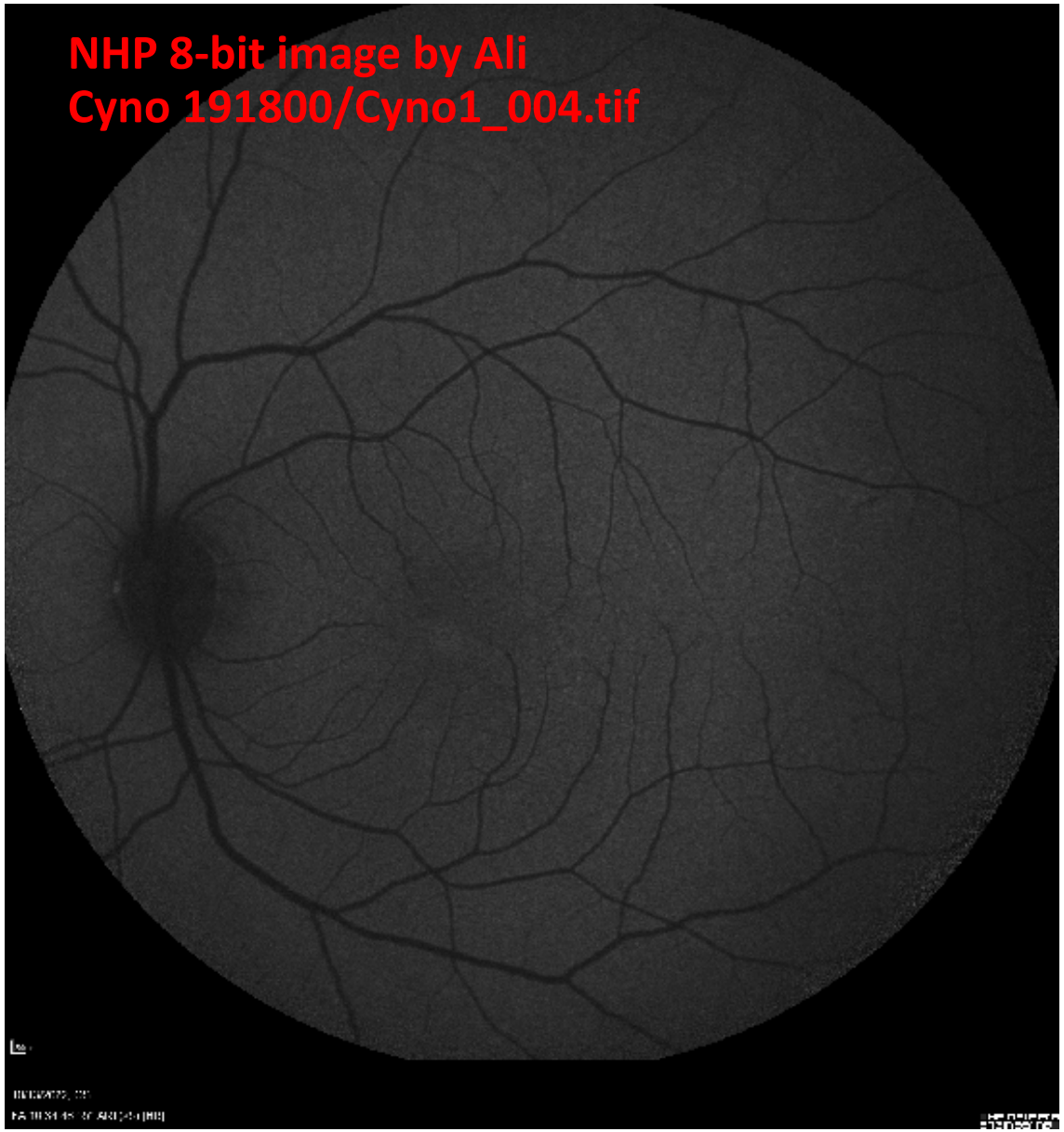


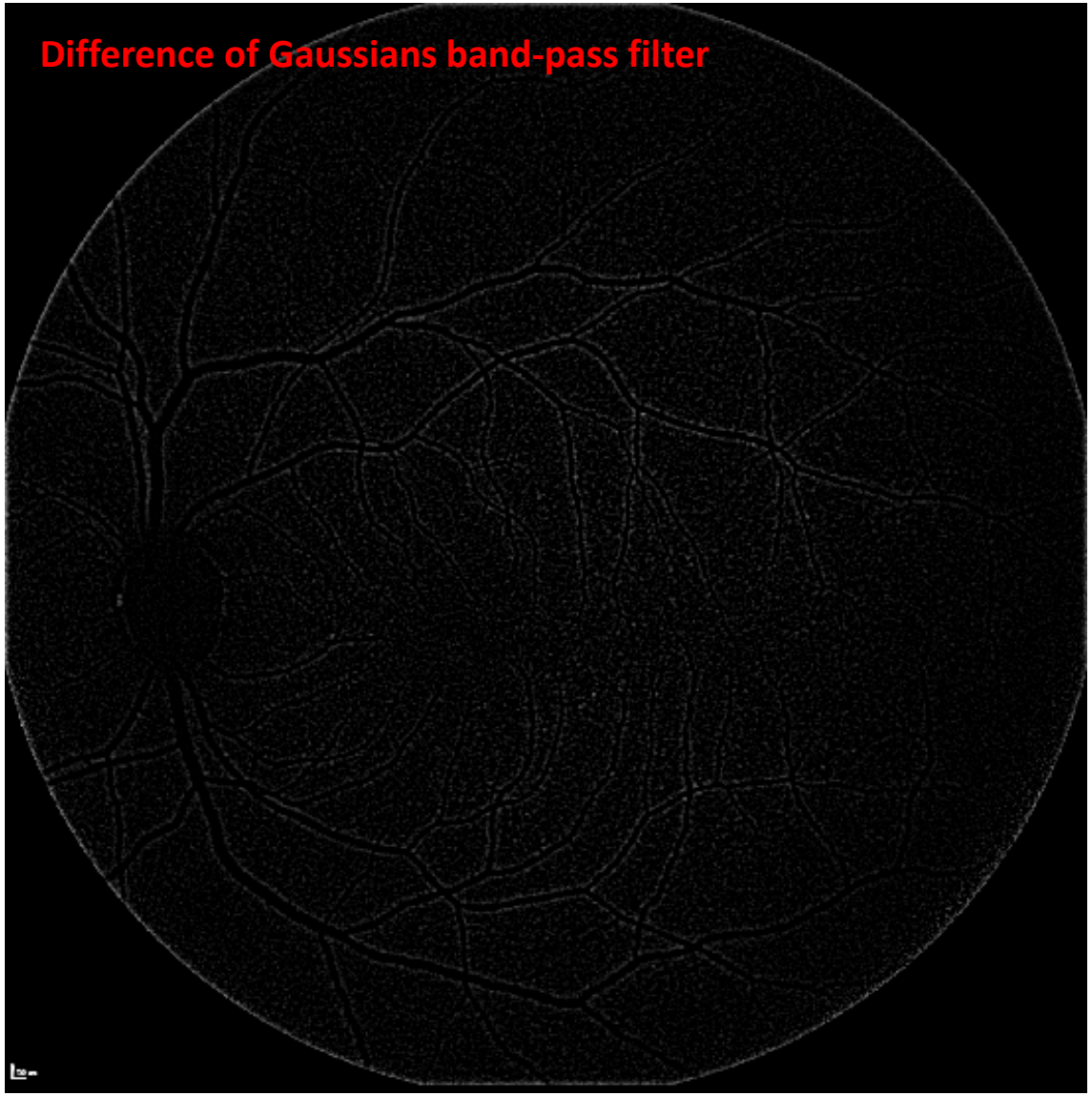
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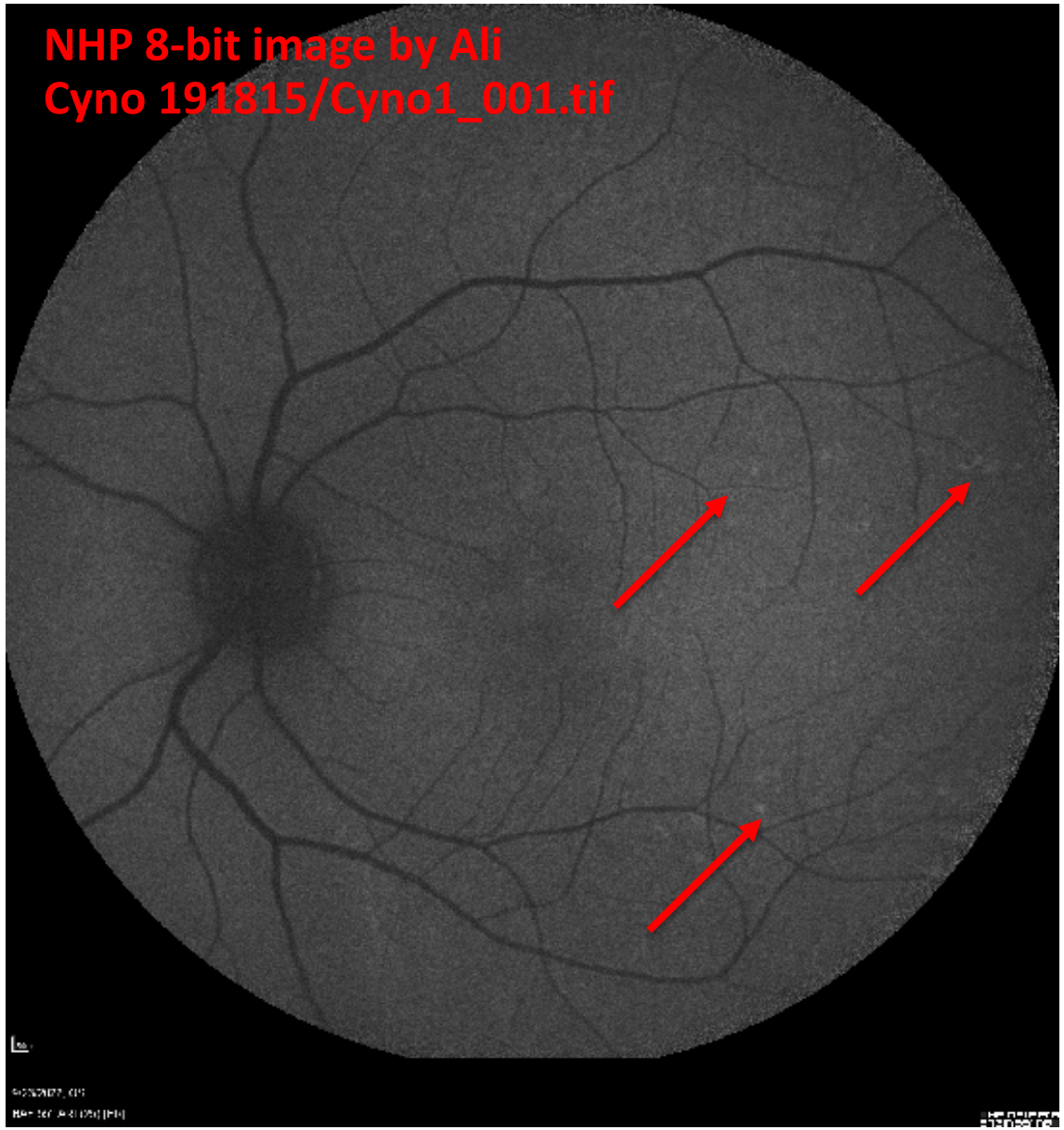


Preliminary analysis



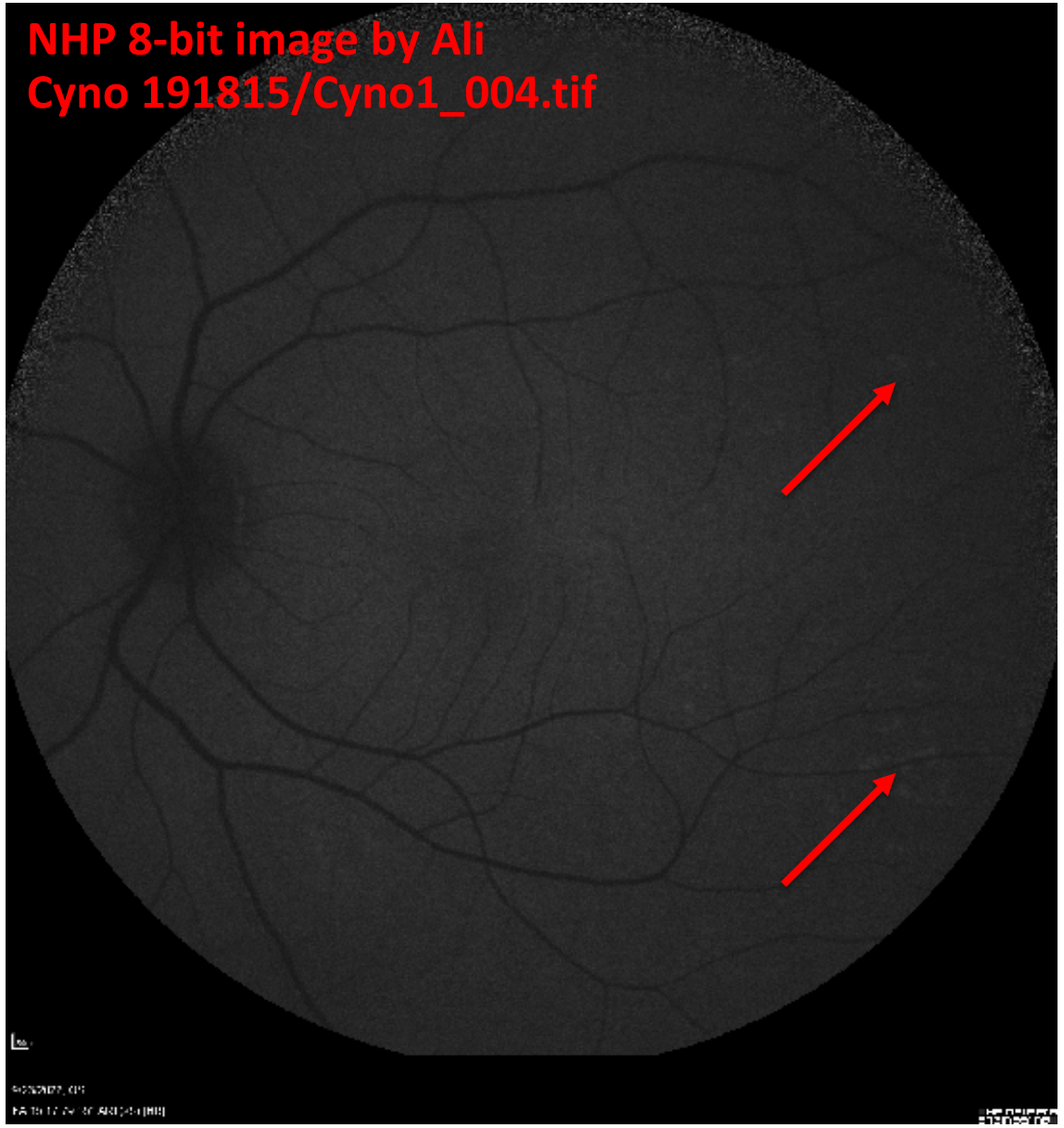


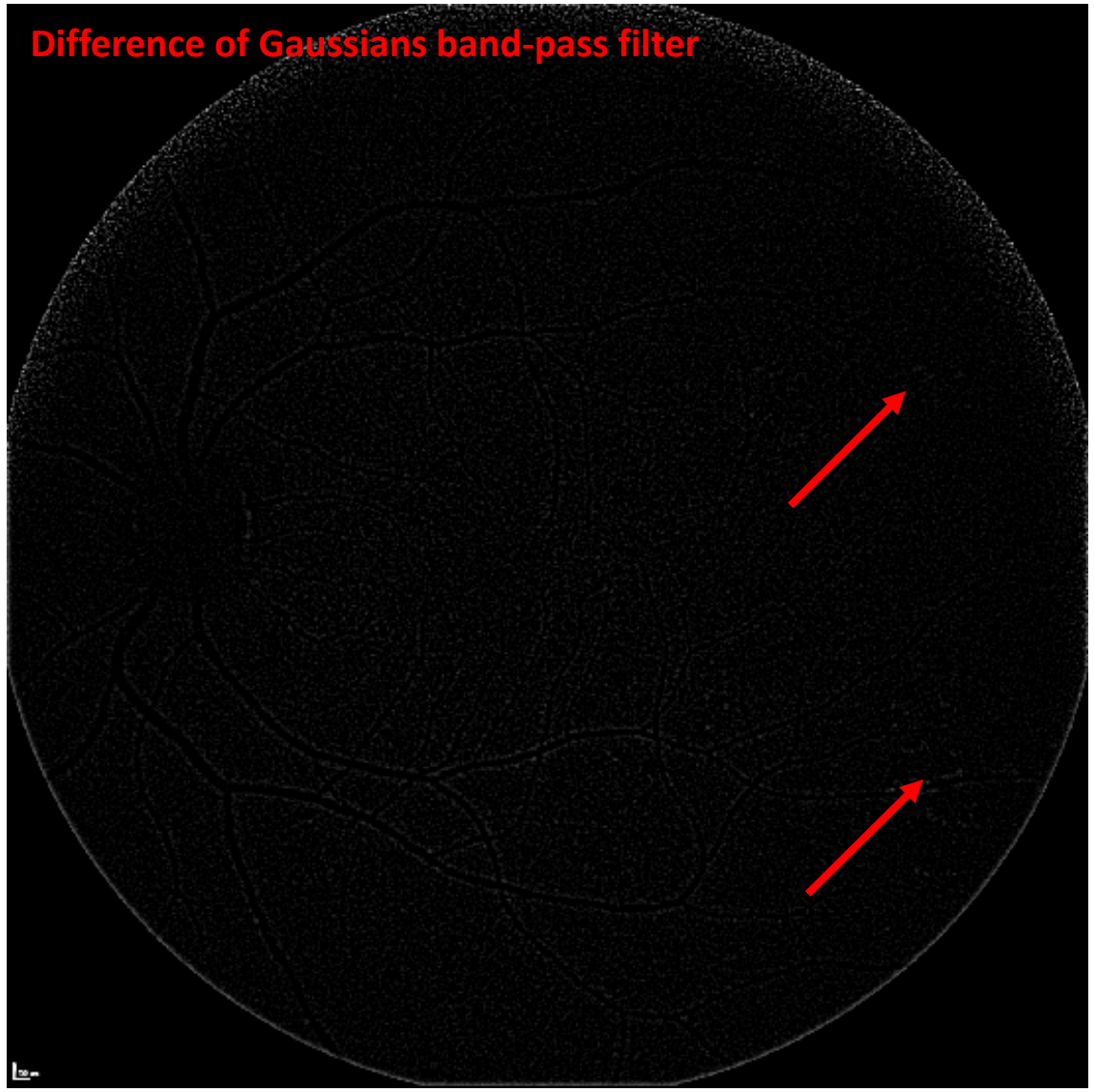
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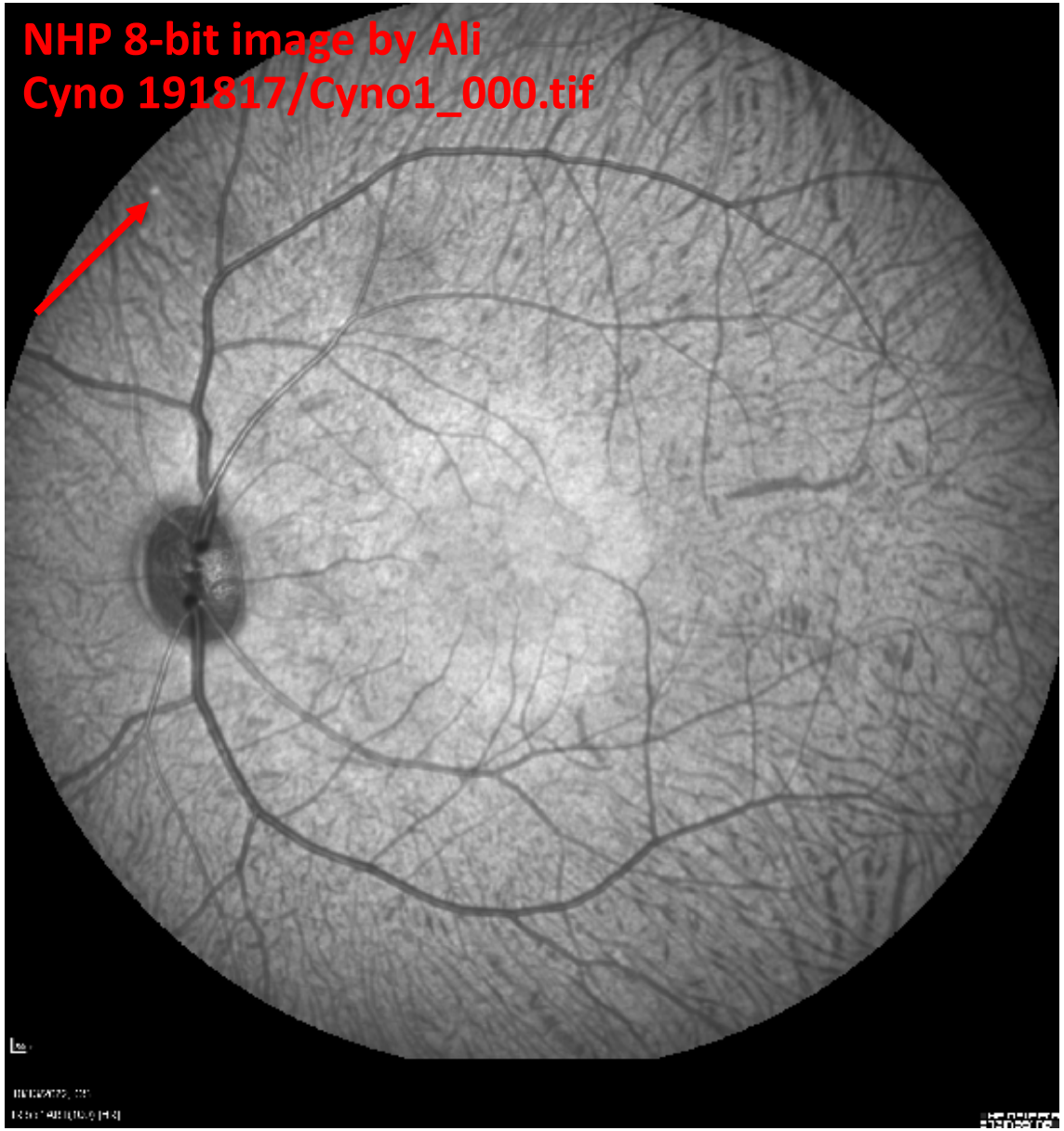


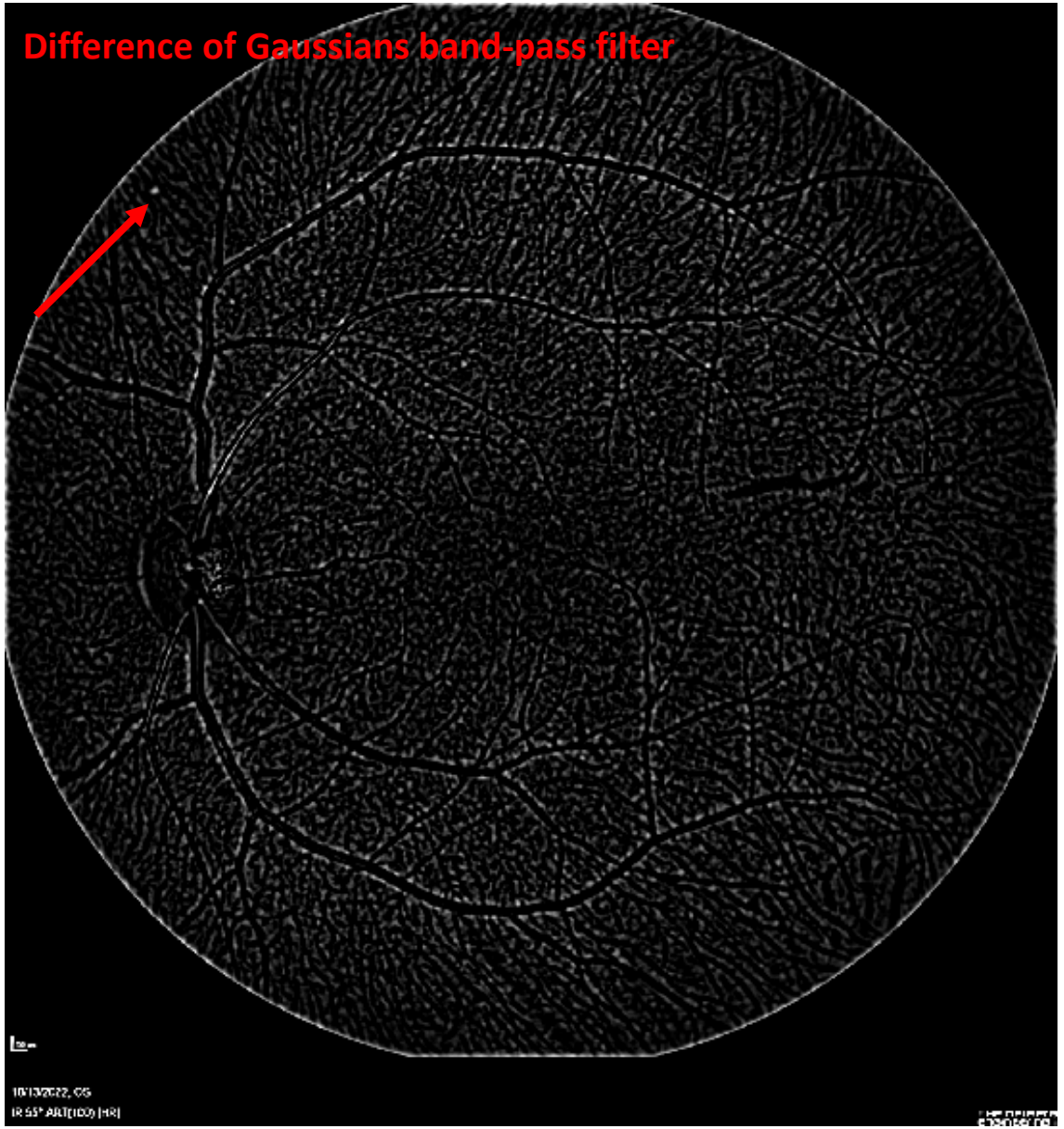
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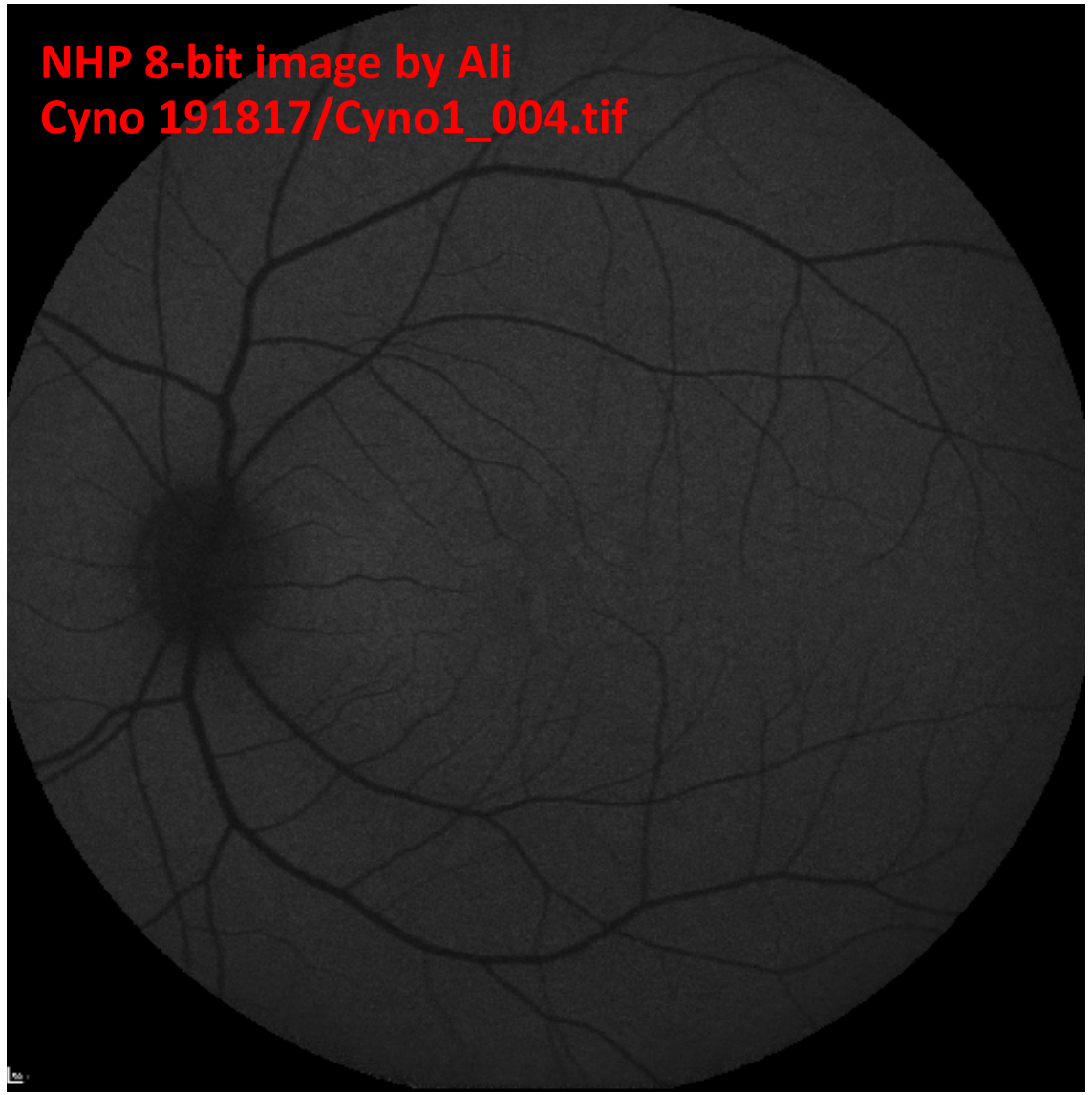


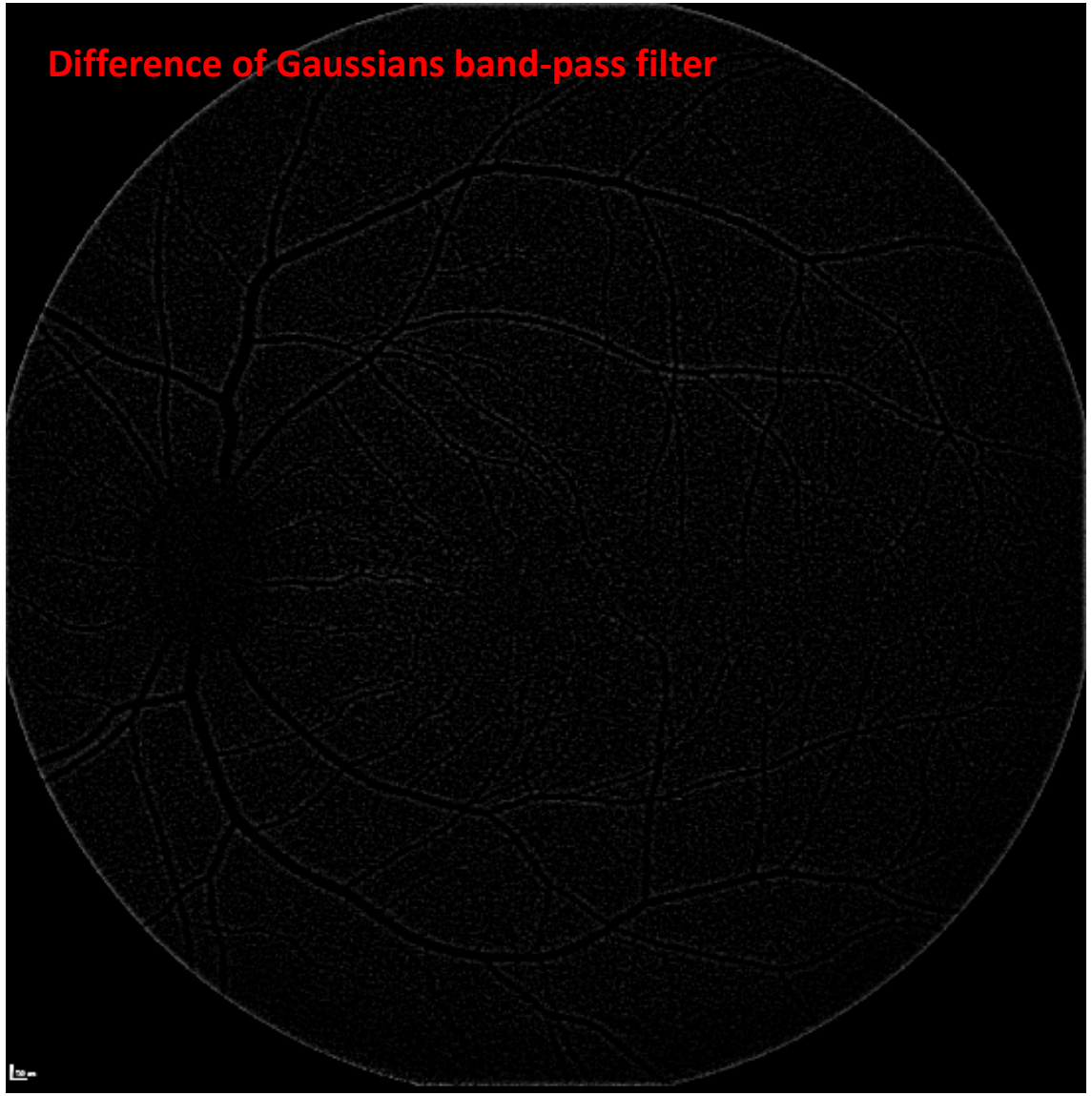
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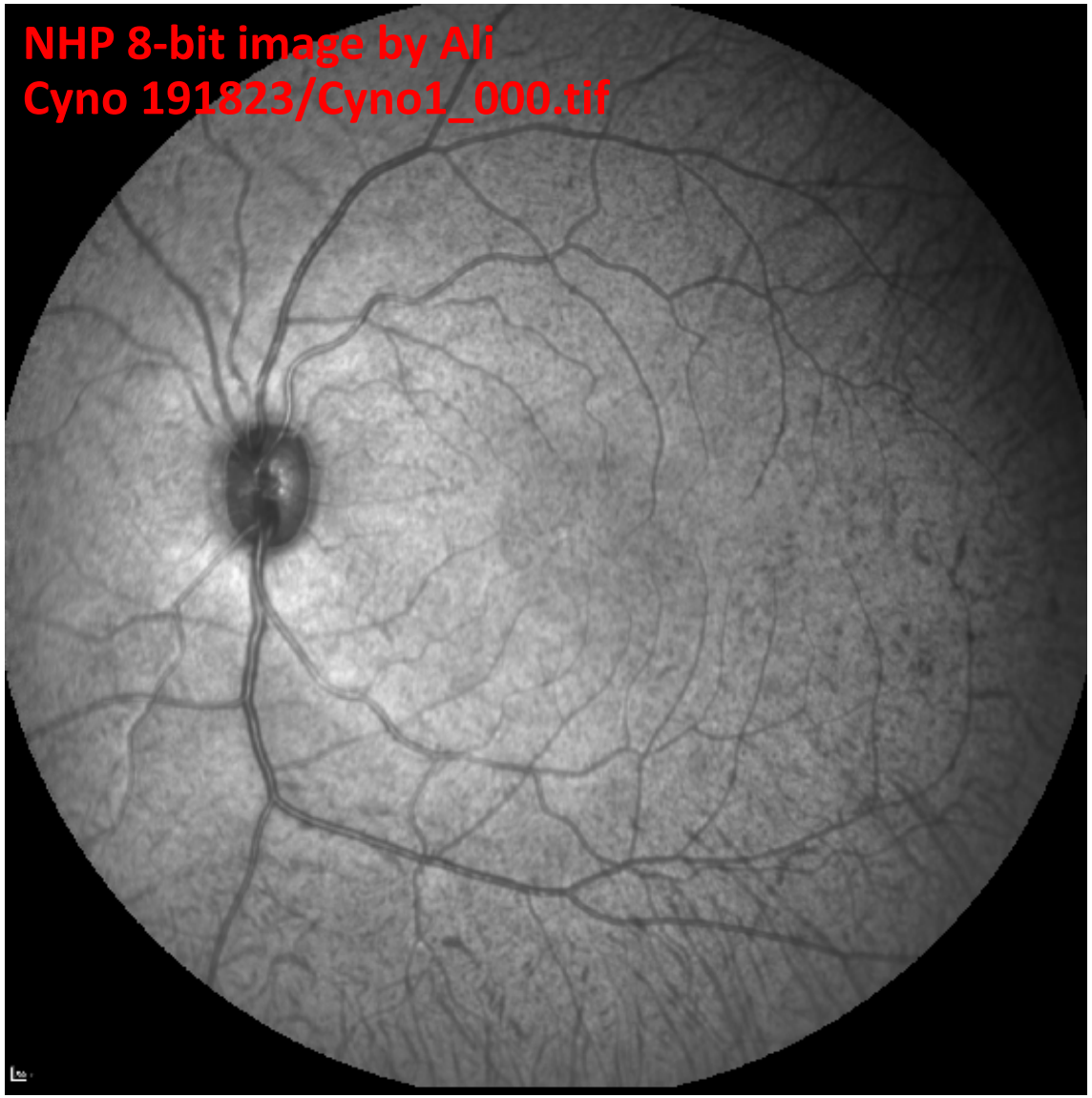


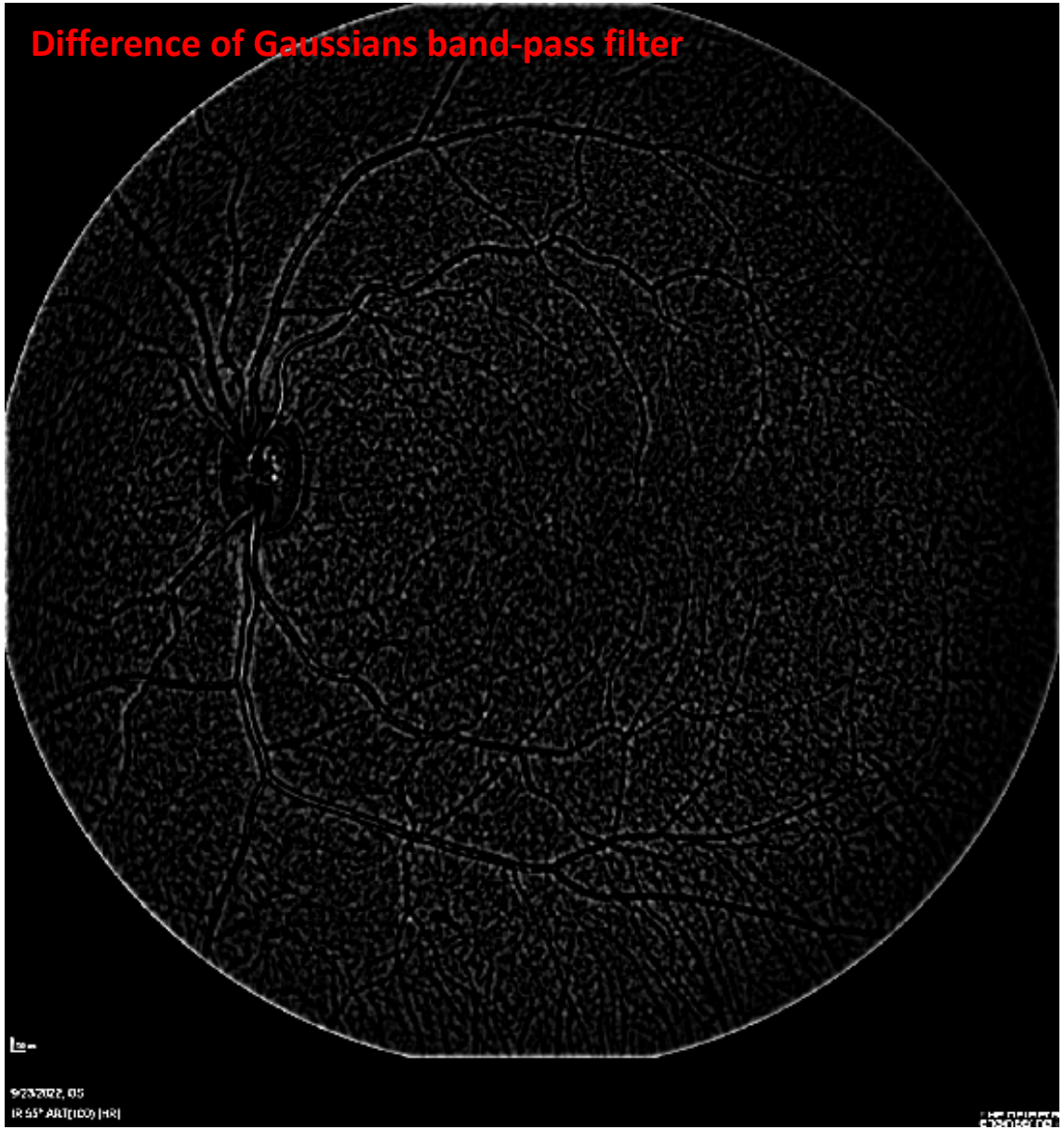
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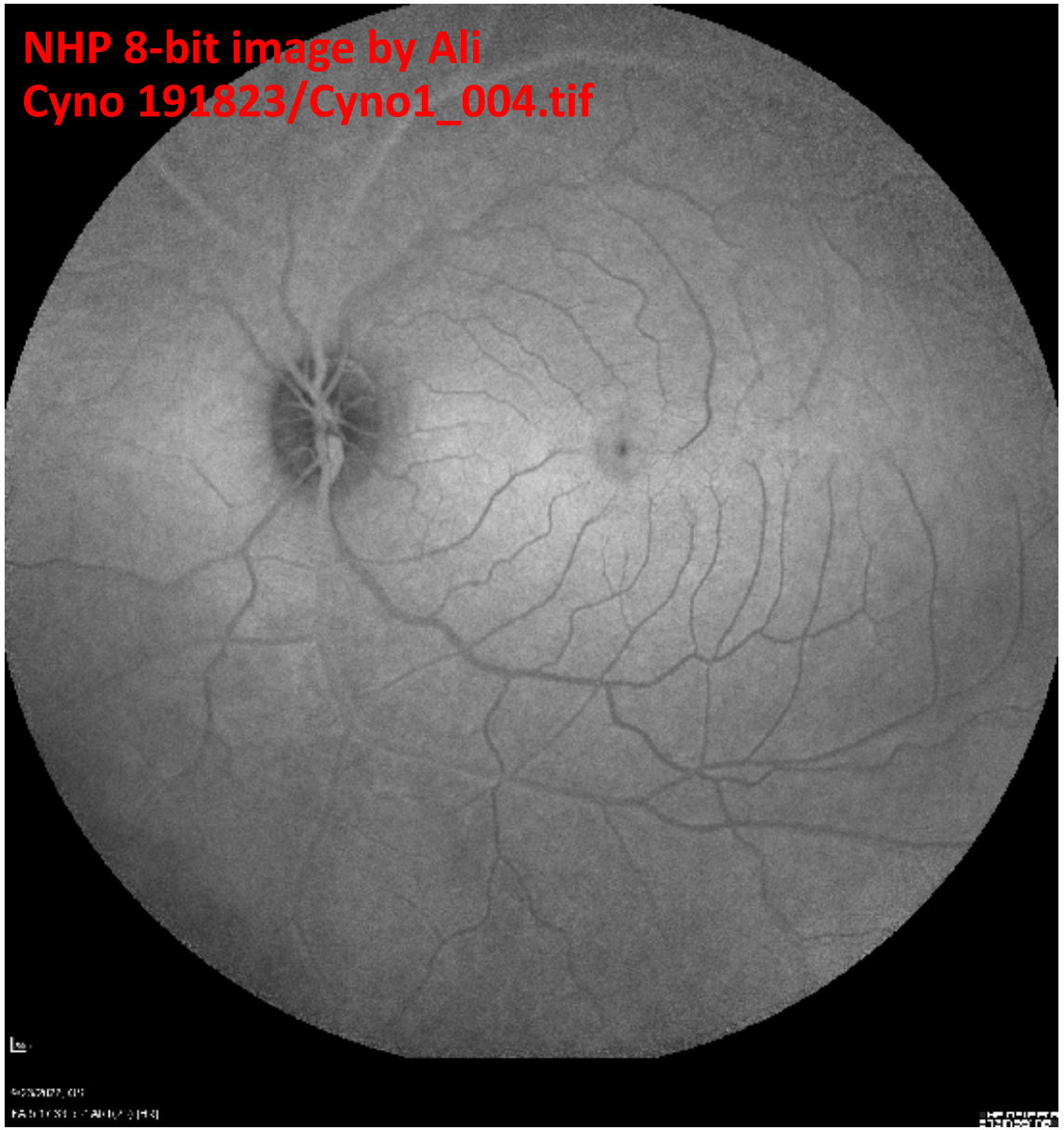


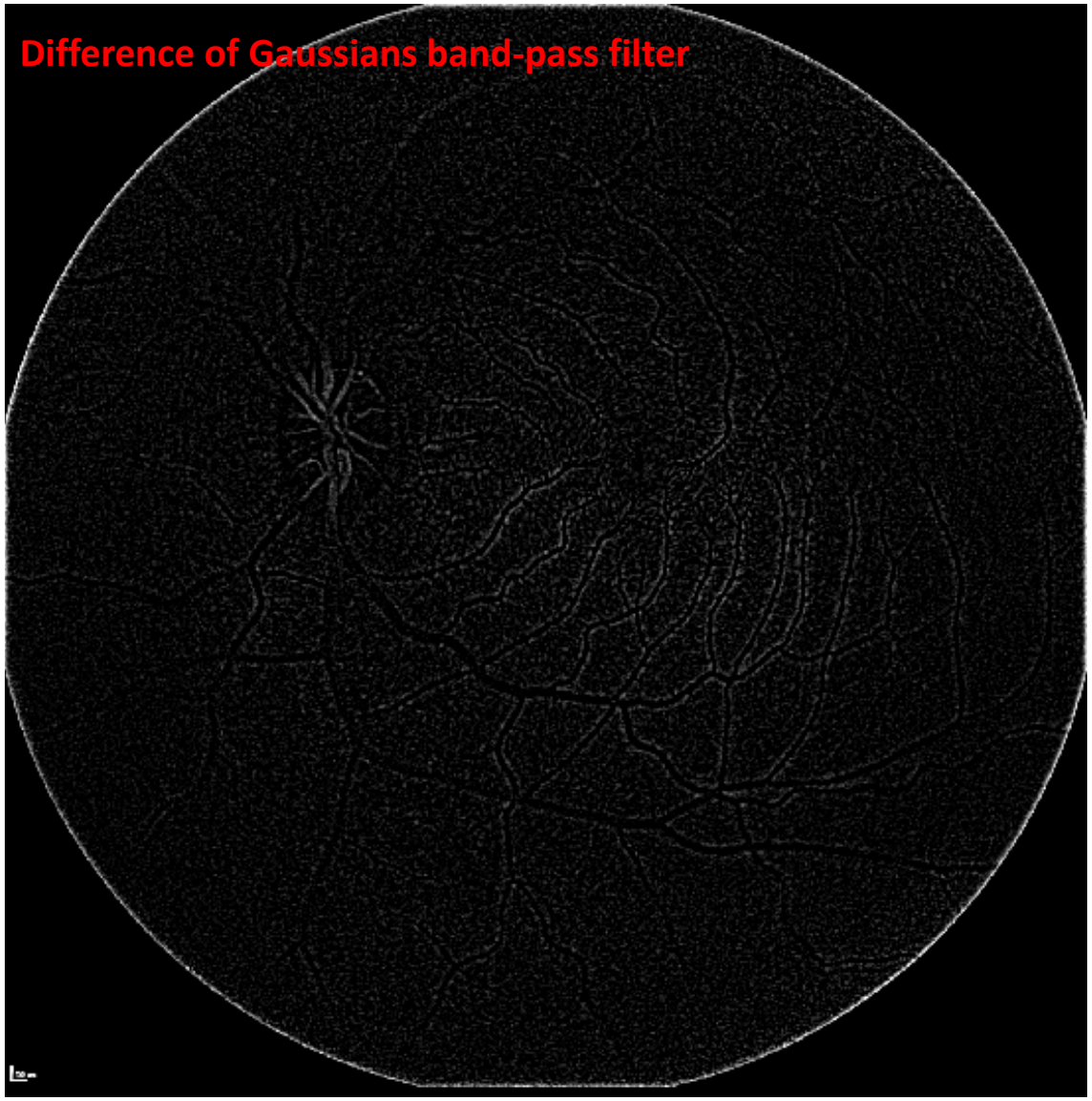
Preliminary analysis



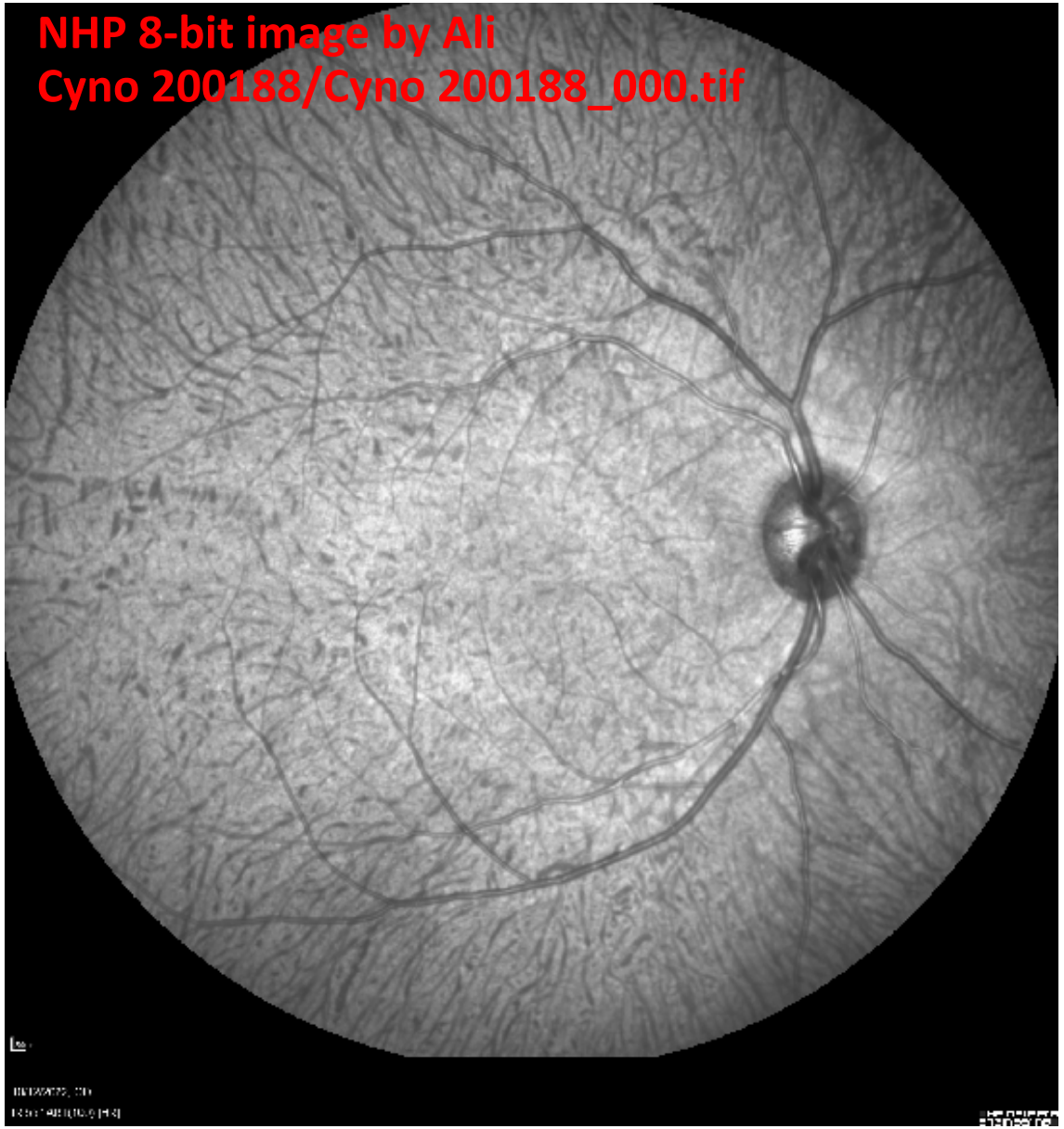


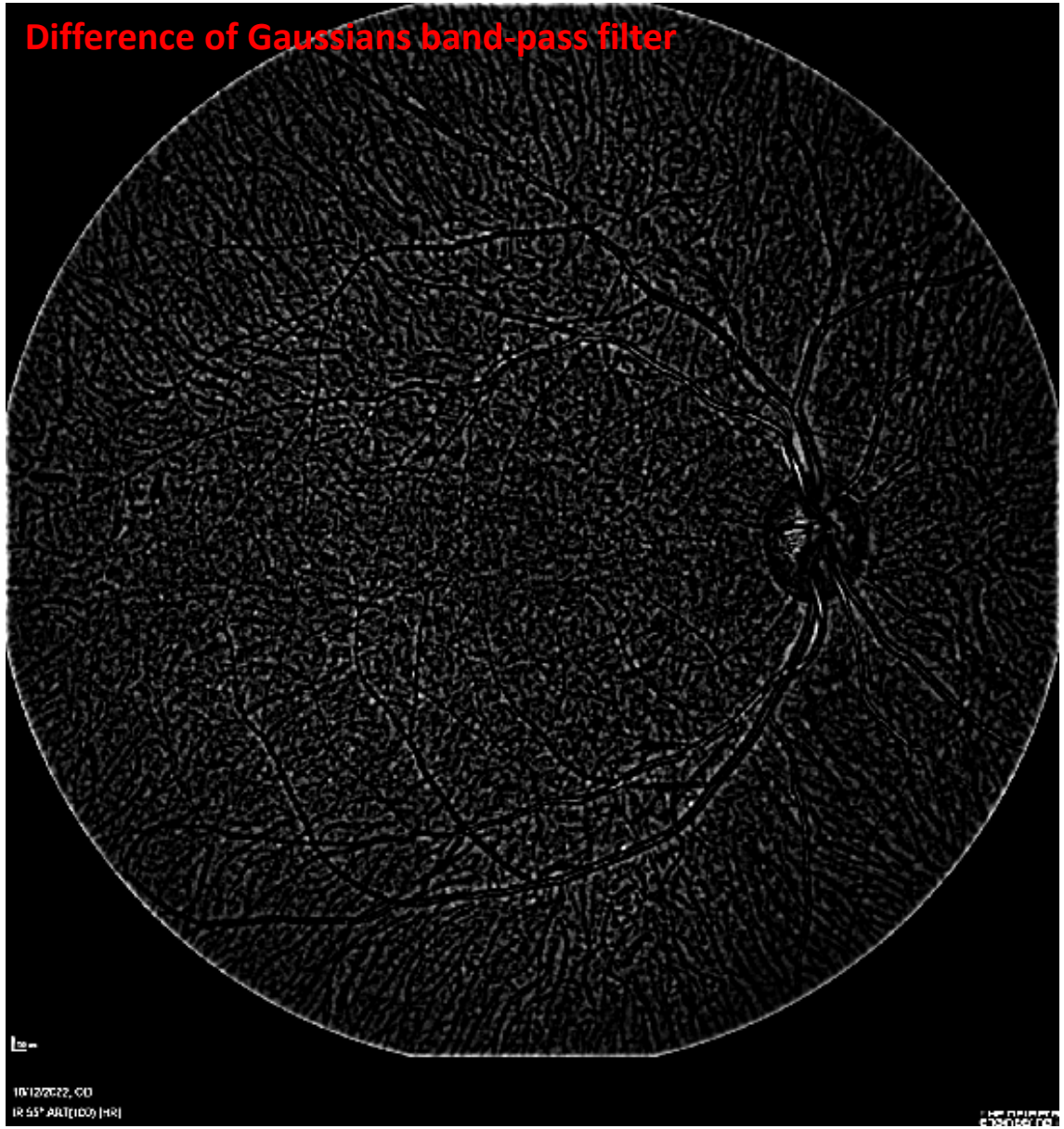
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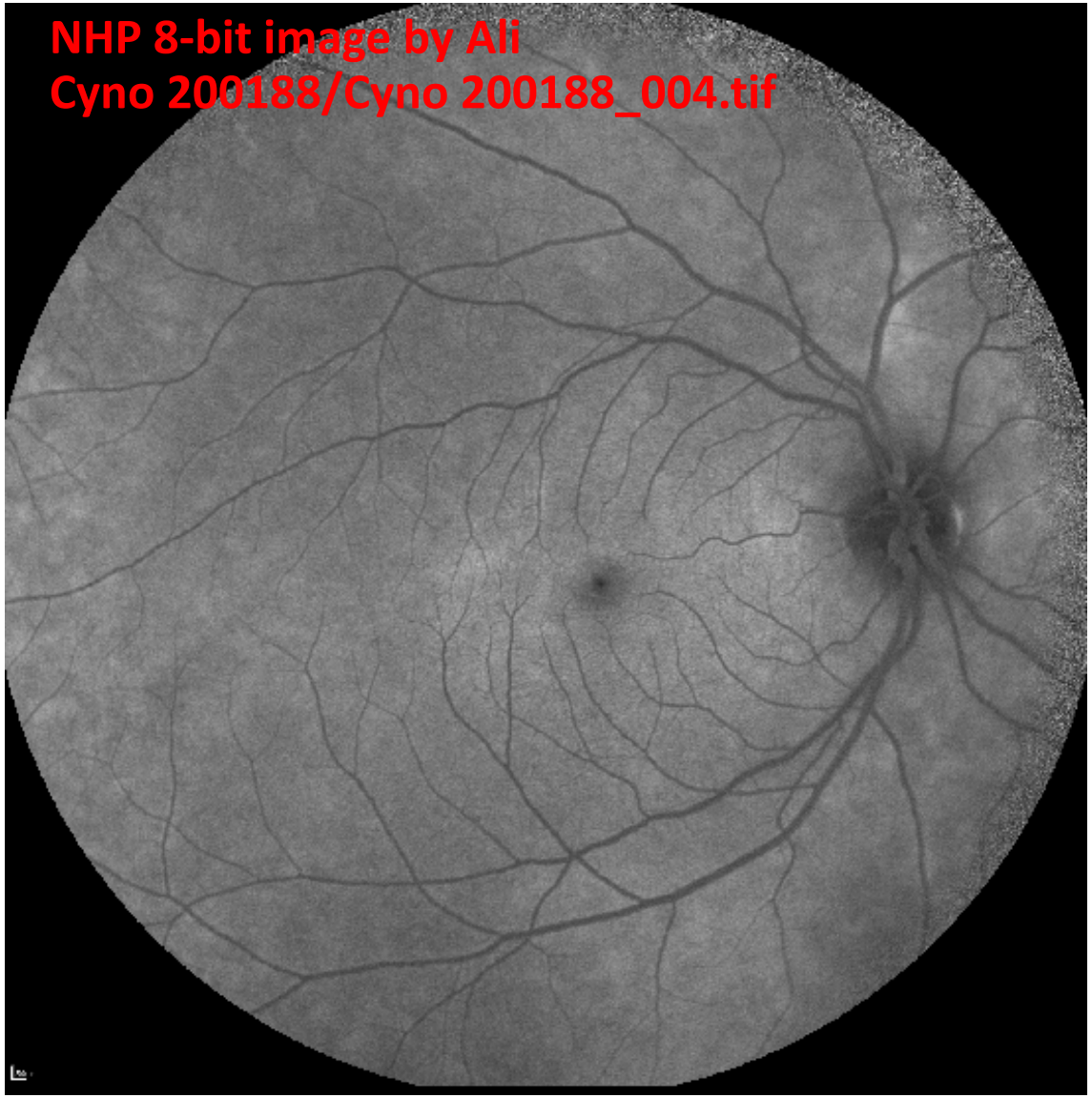


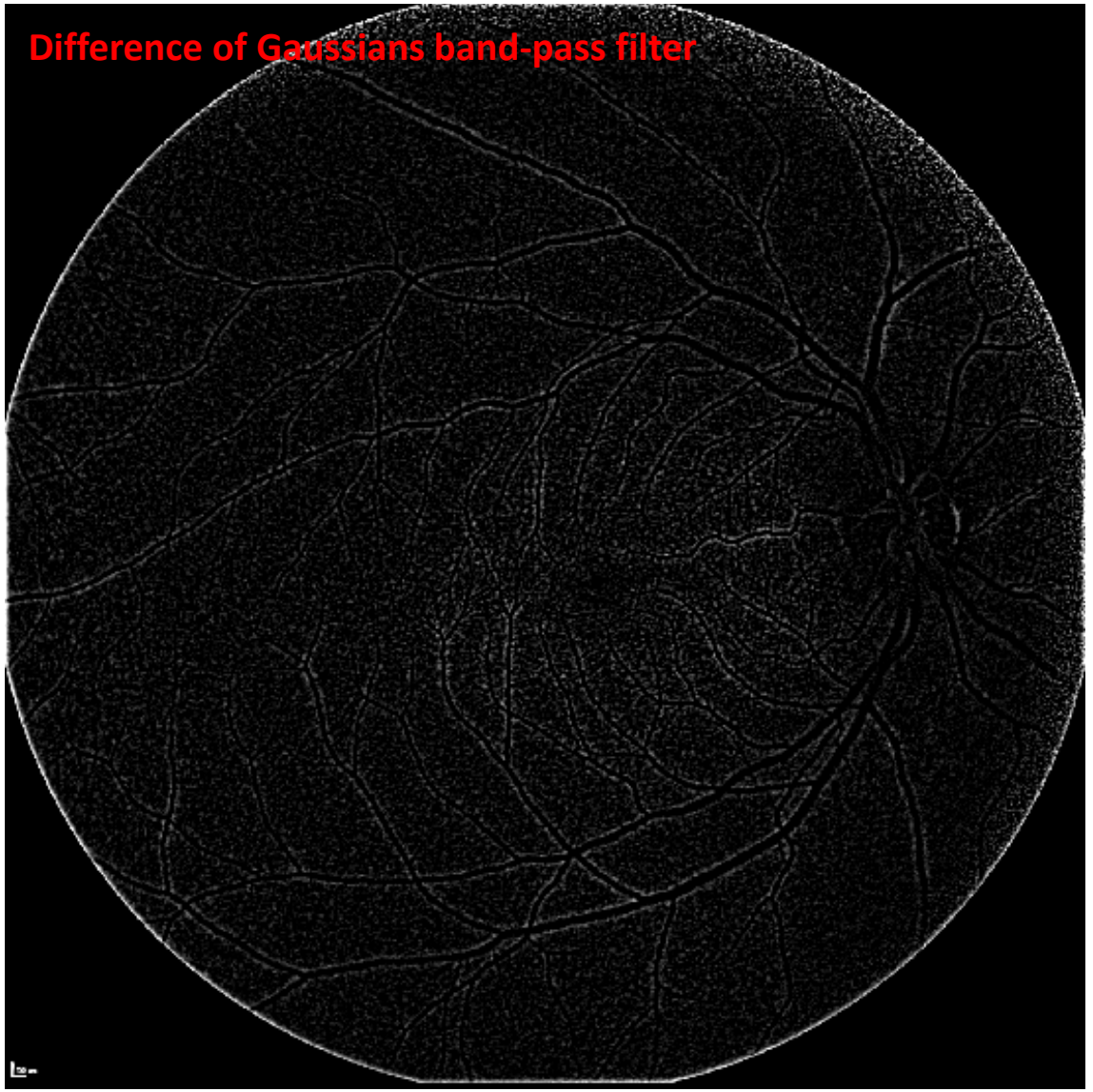
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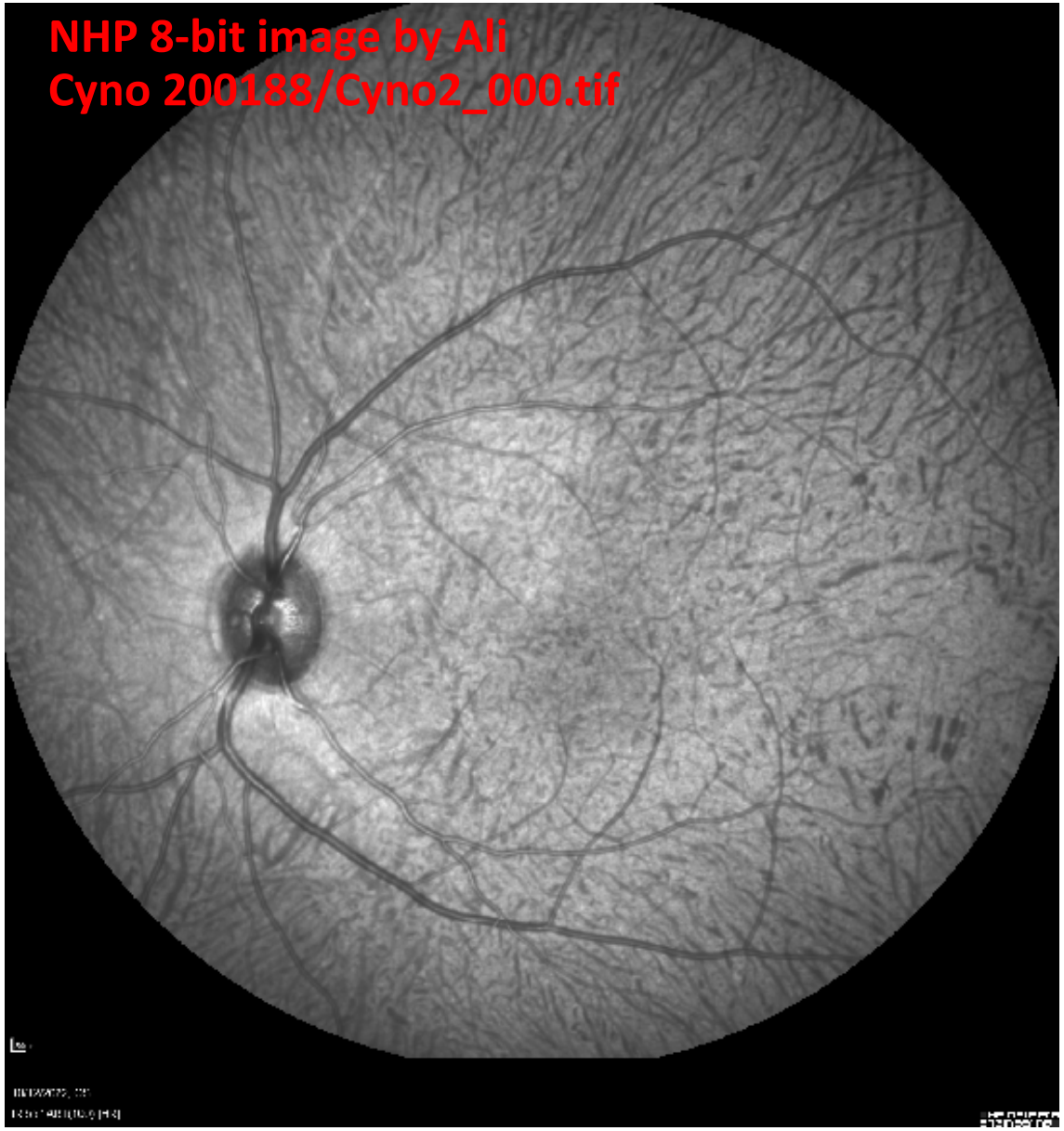


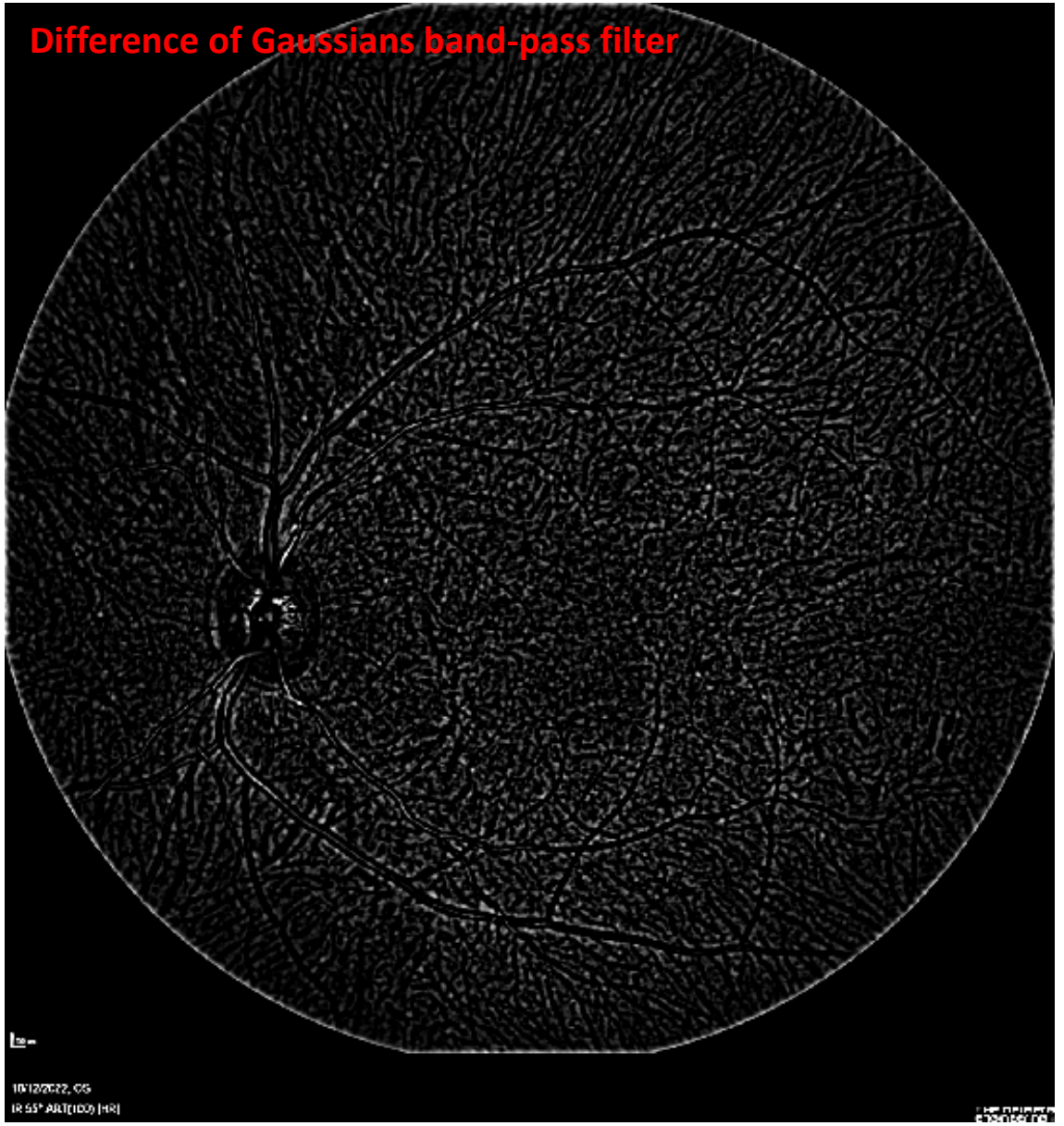
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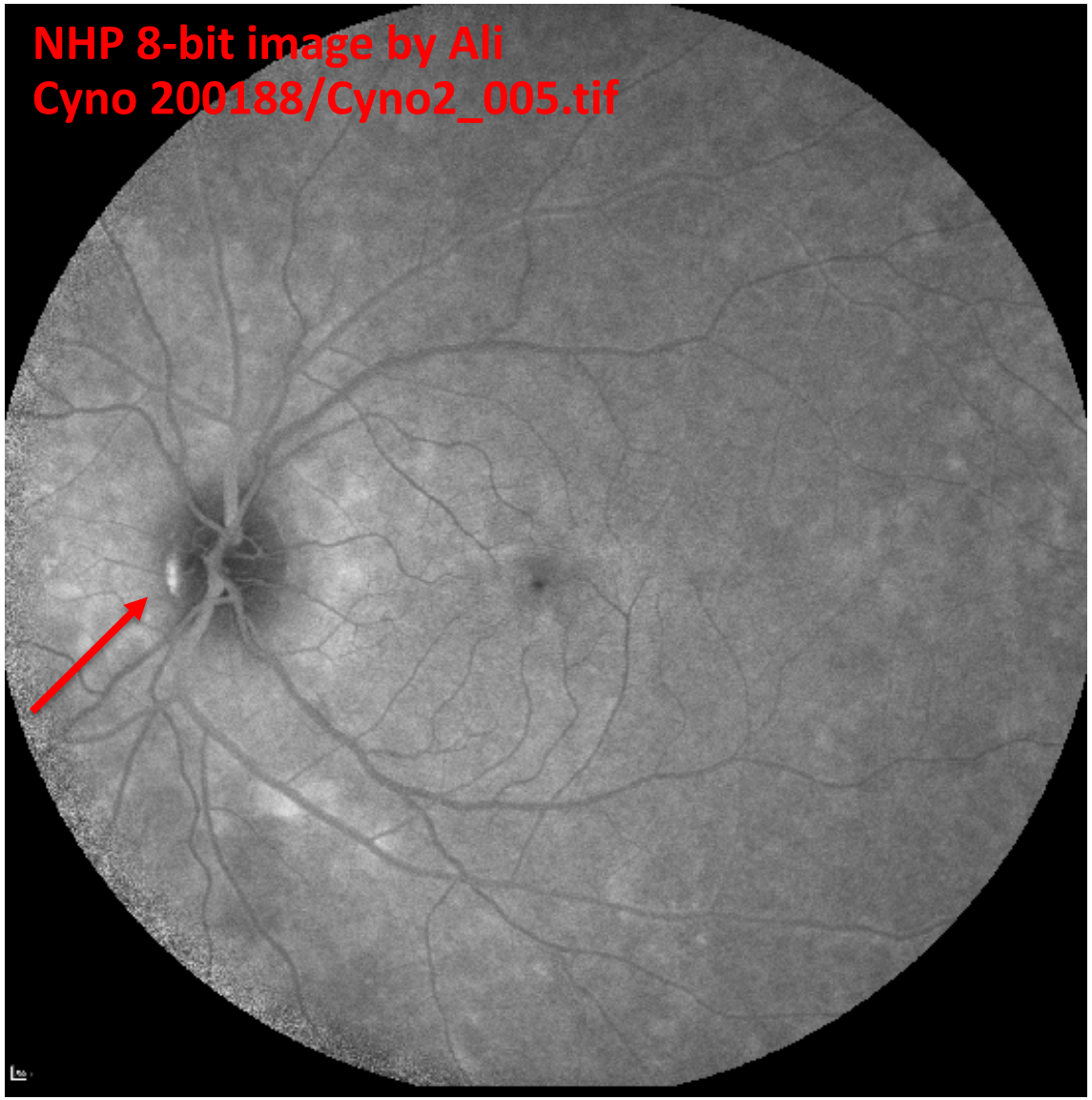


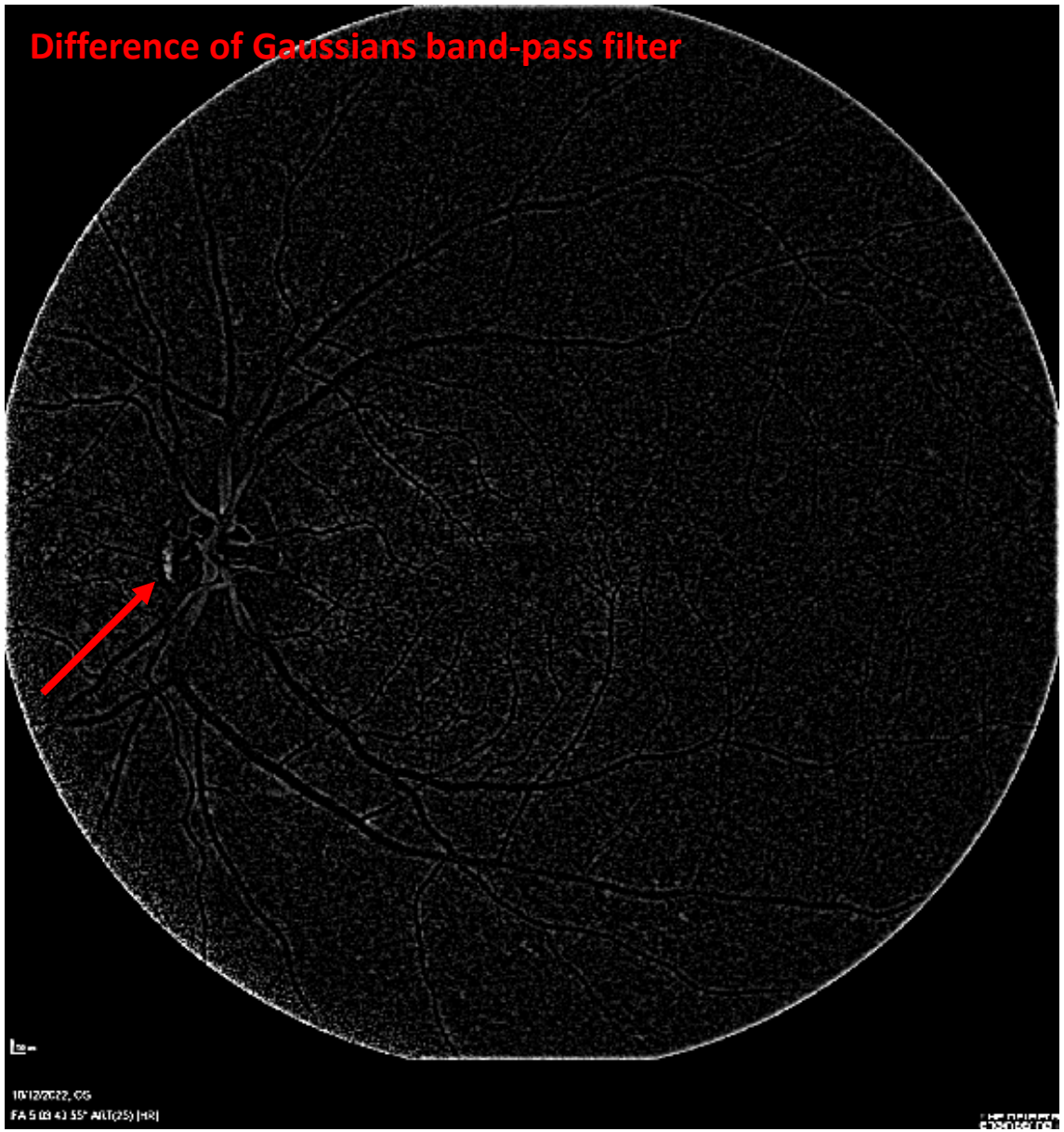
Preliminary analysis



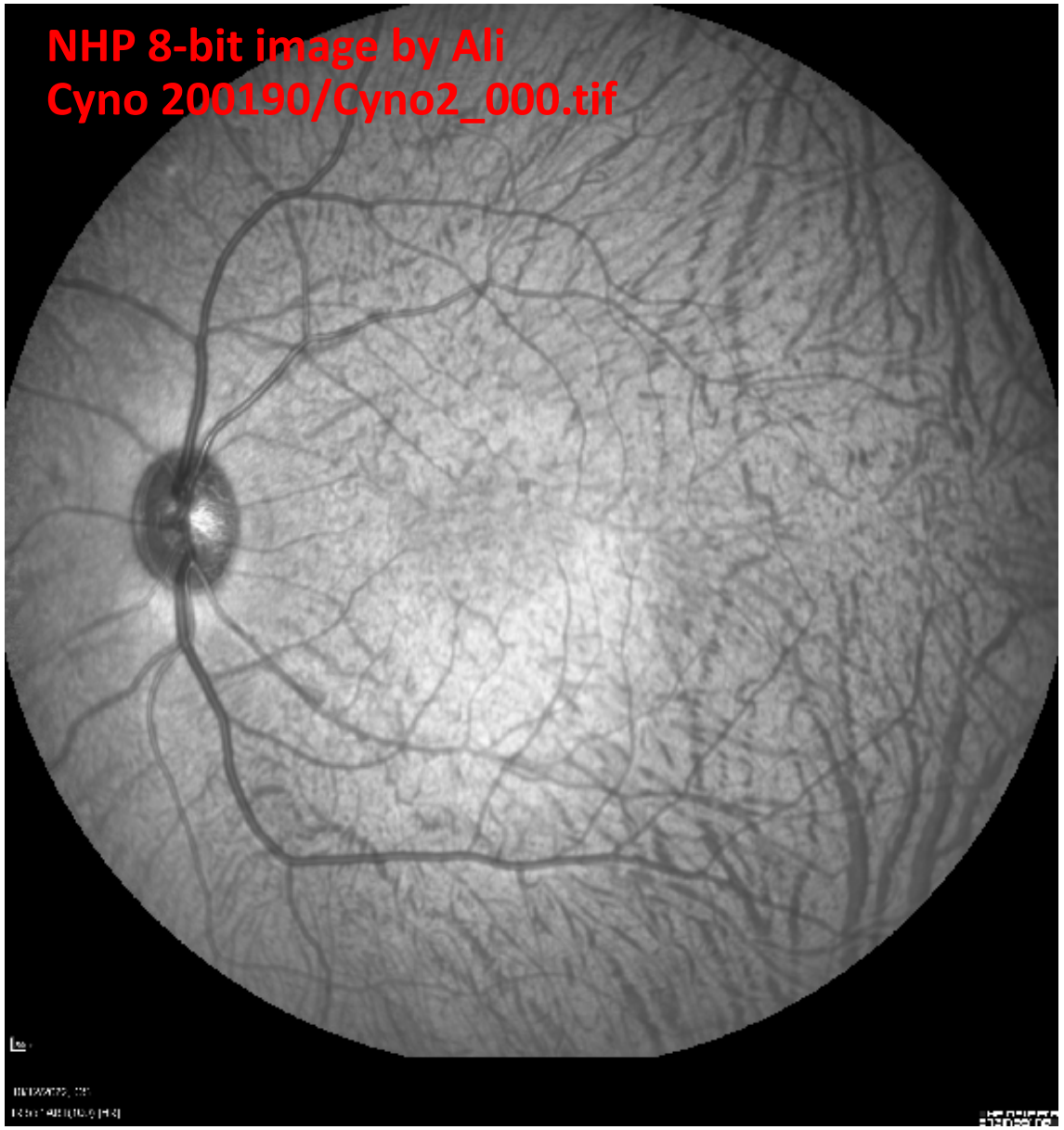


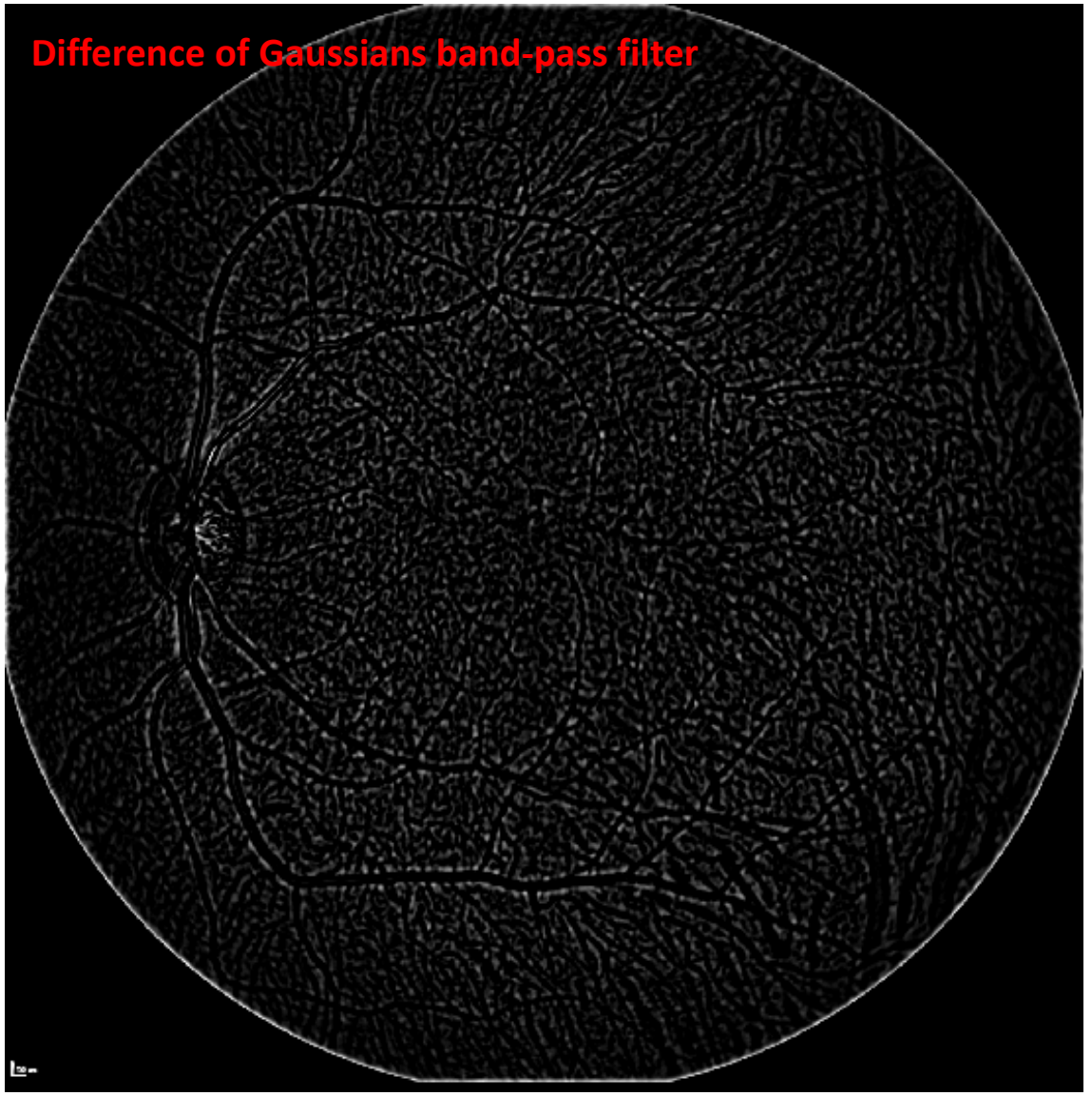
Preliminary analysis



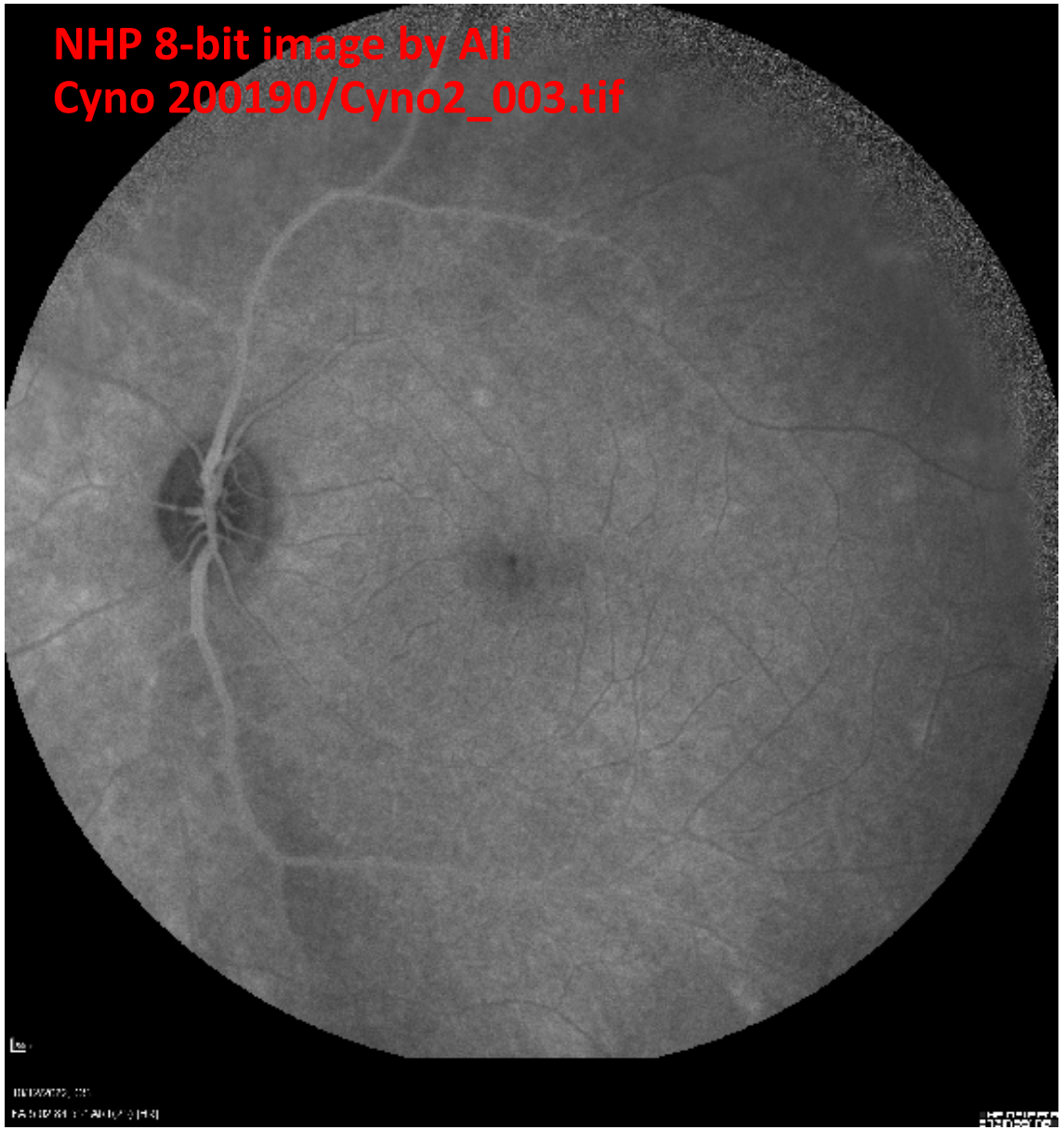


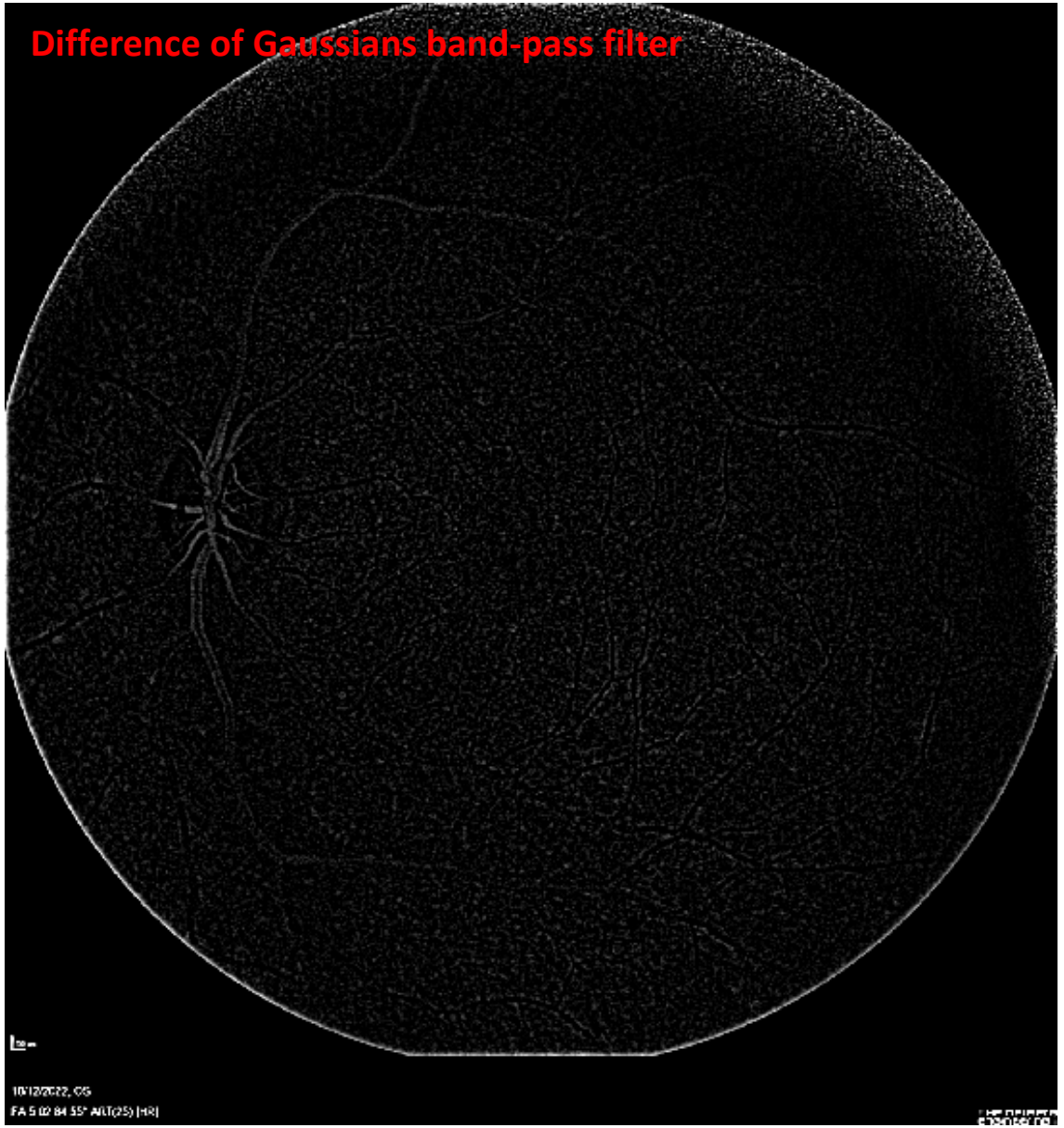
Preliminary analysis





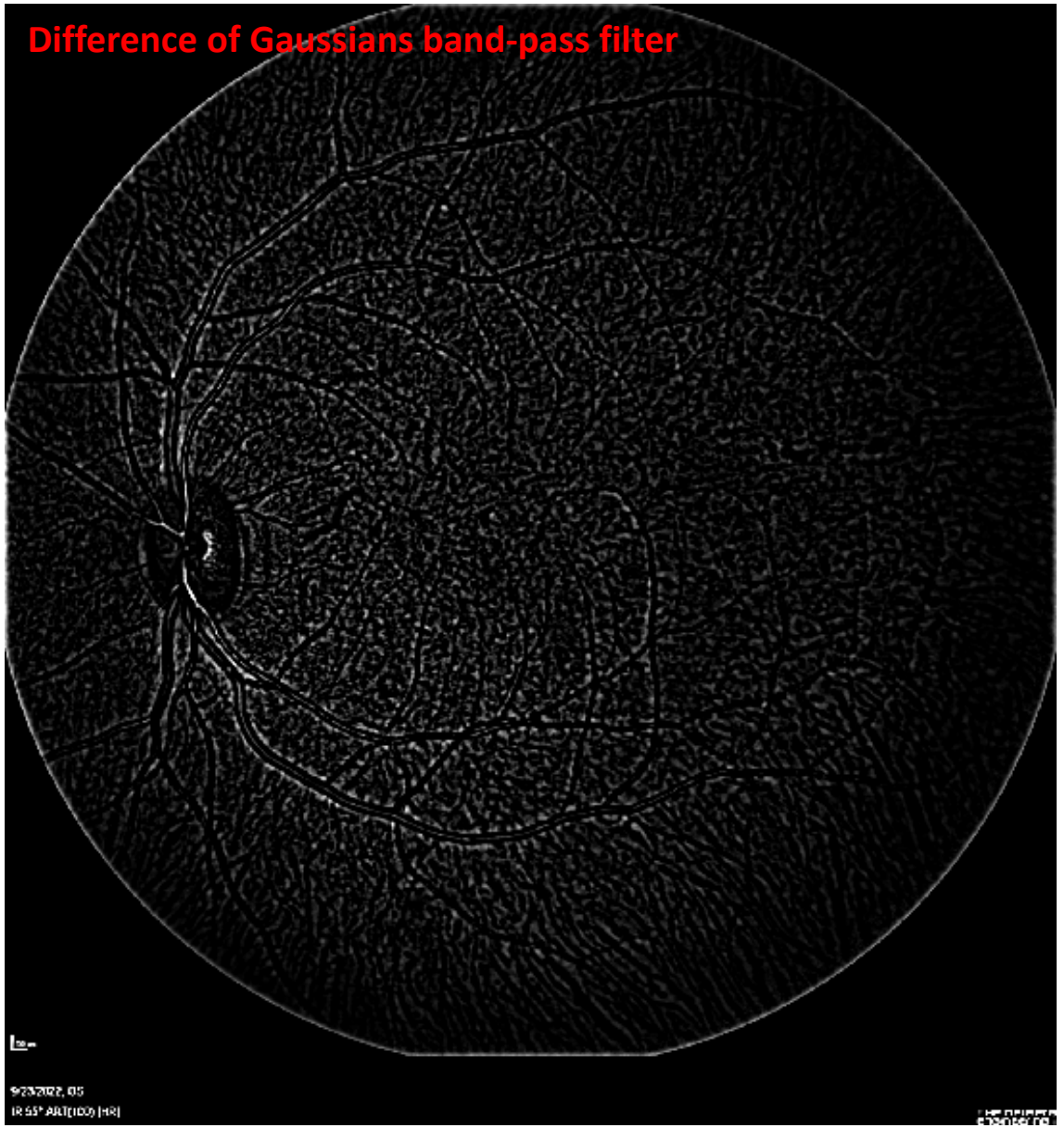
Preliminary analysis



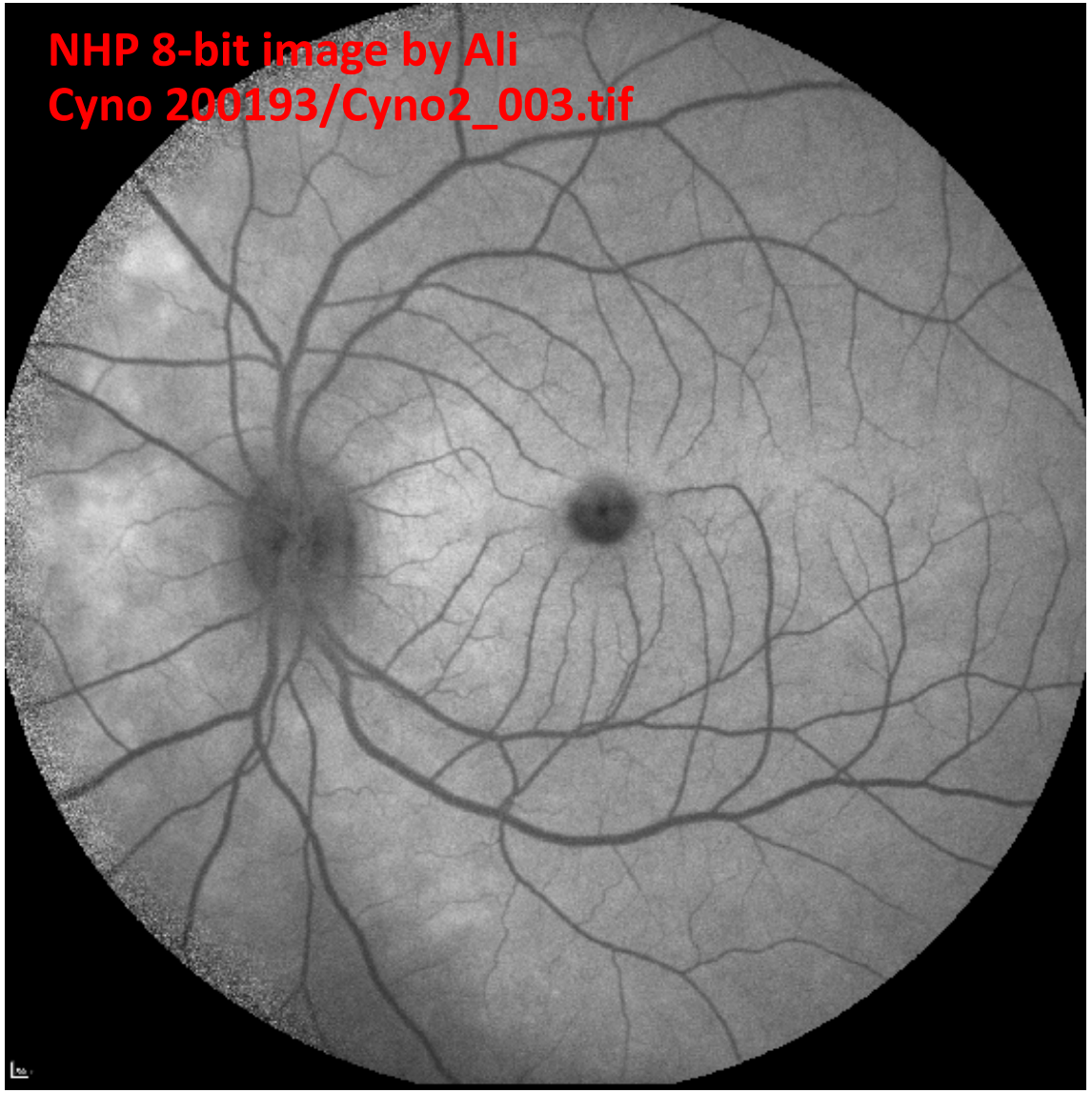


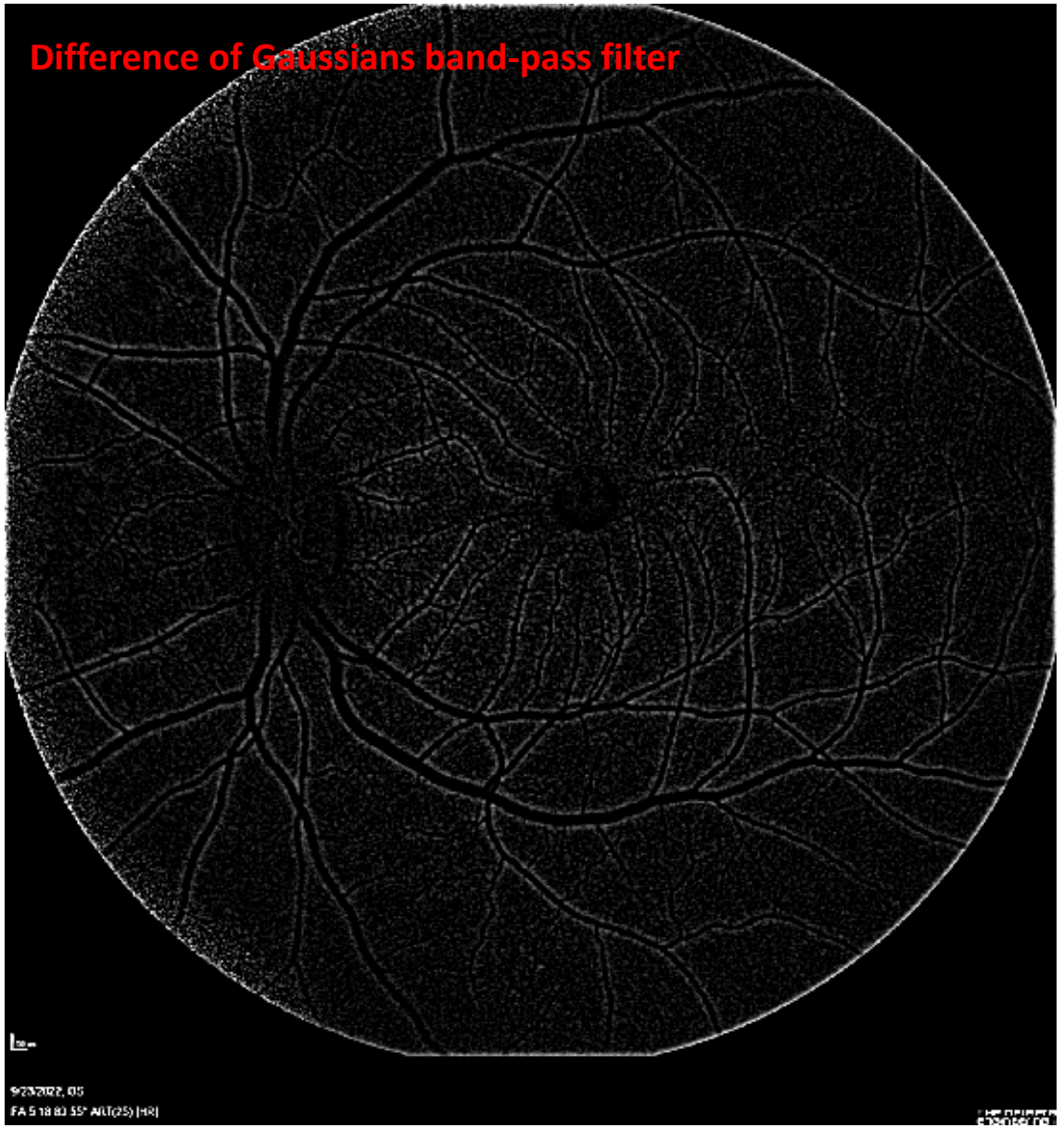
Preliminary analysis



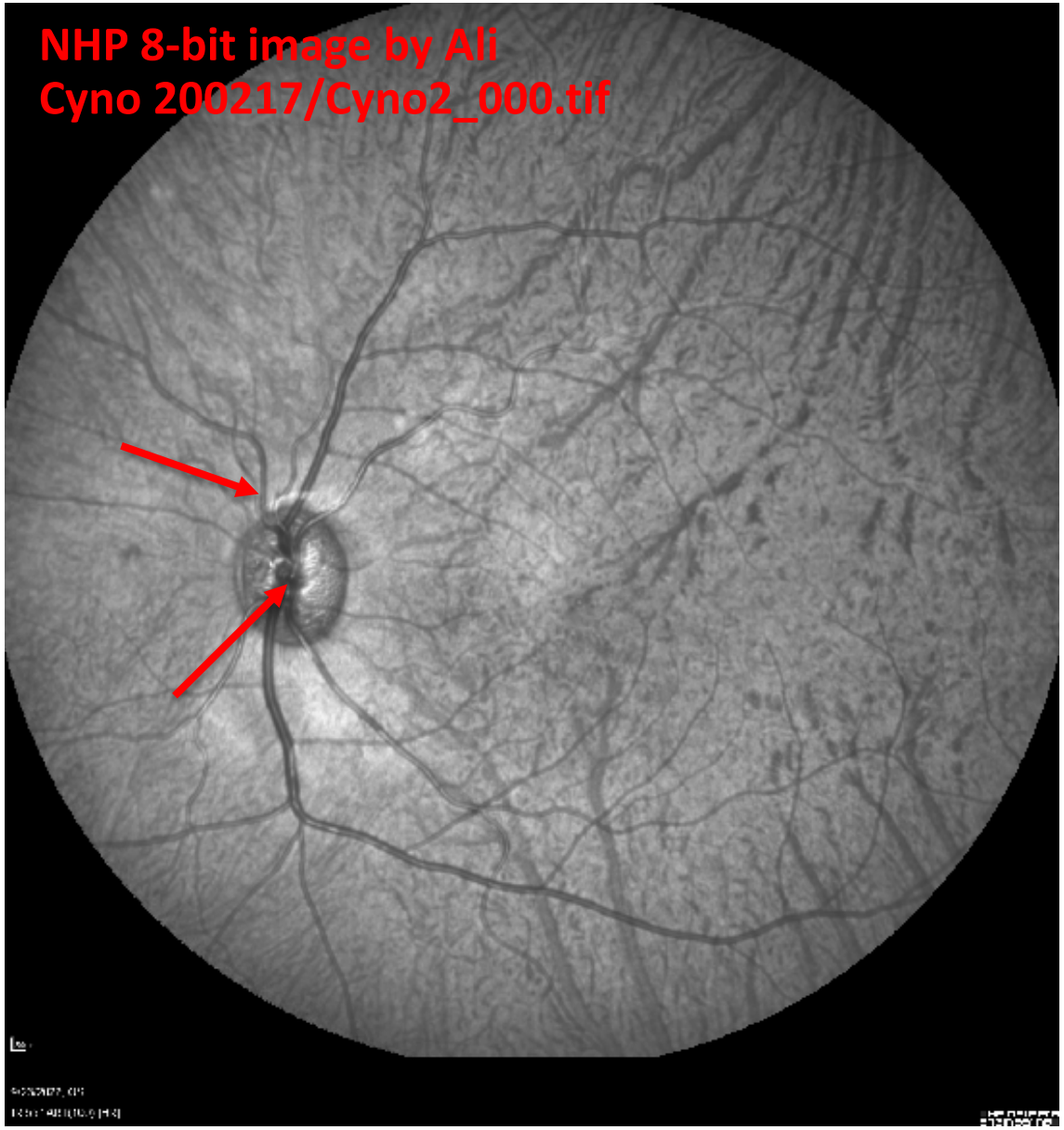


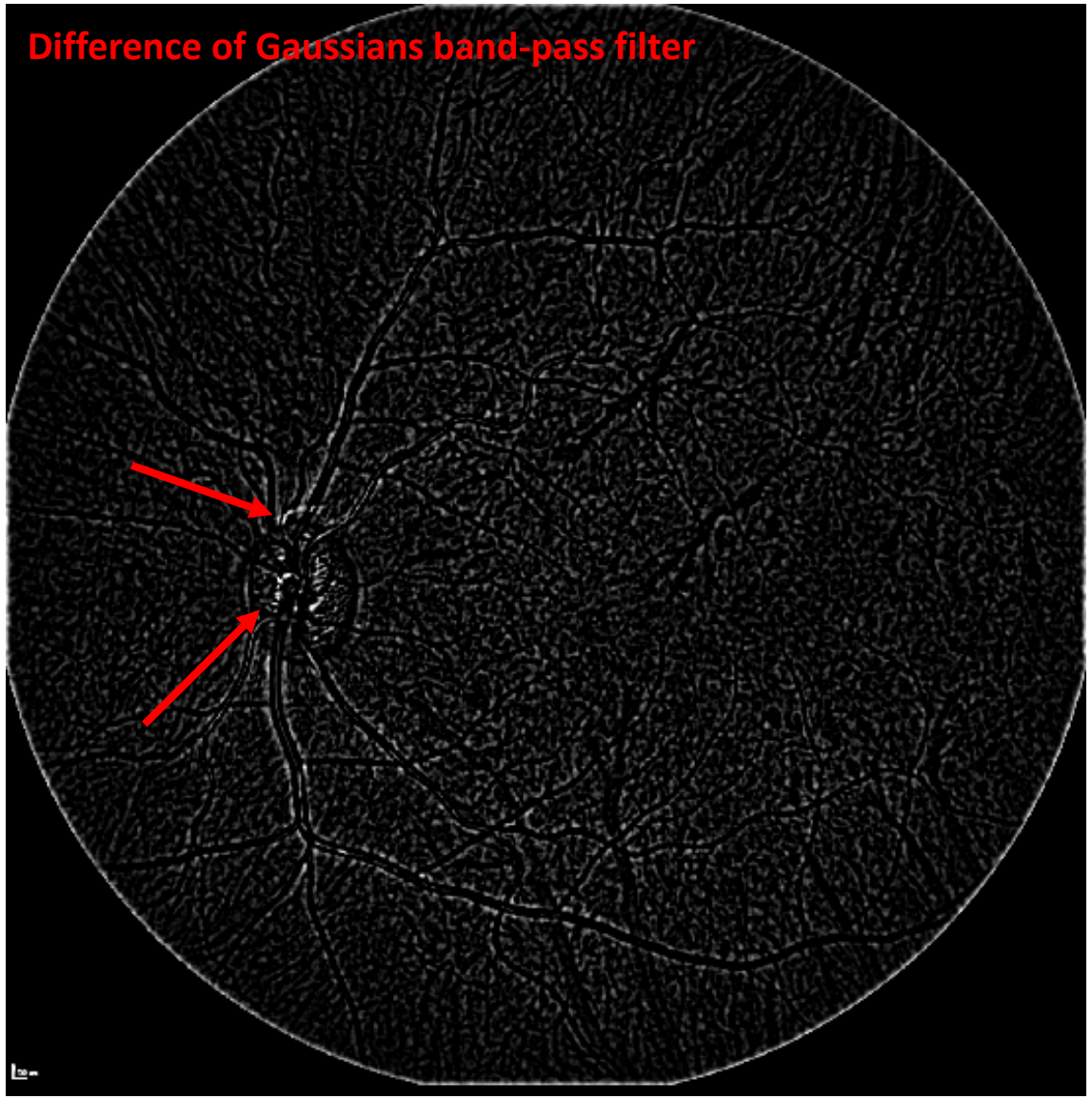
Preliminary analysis



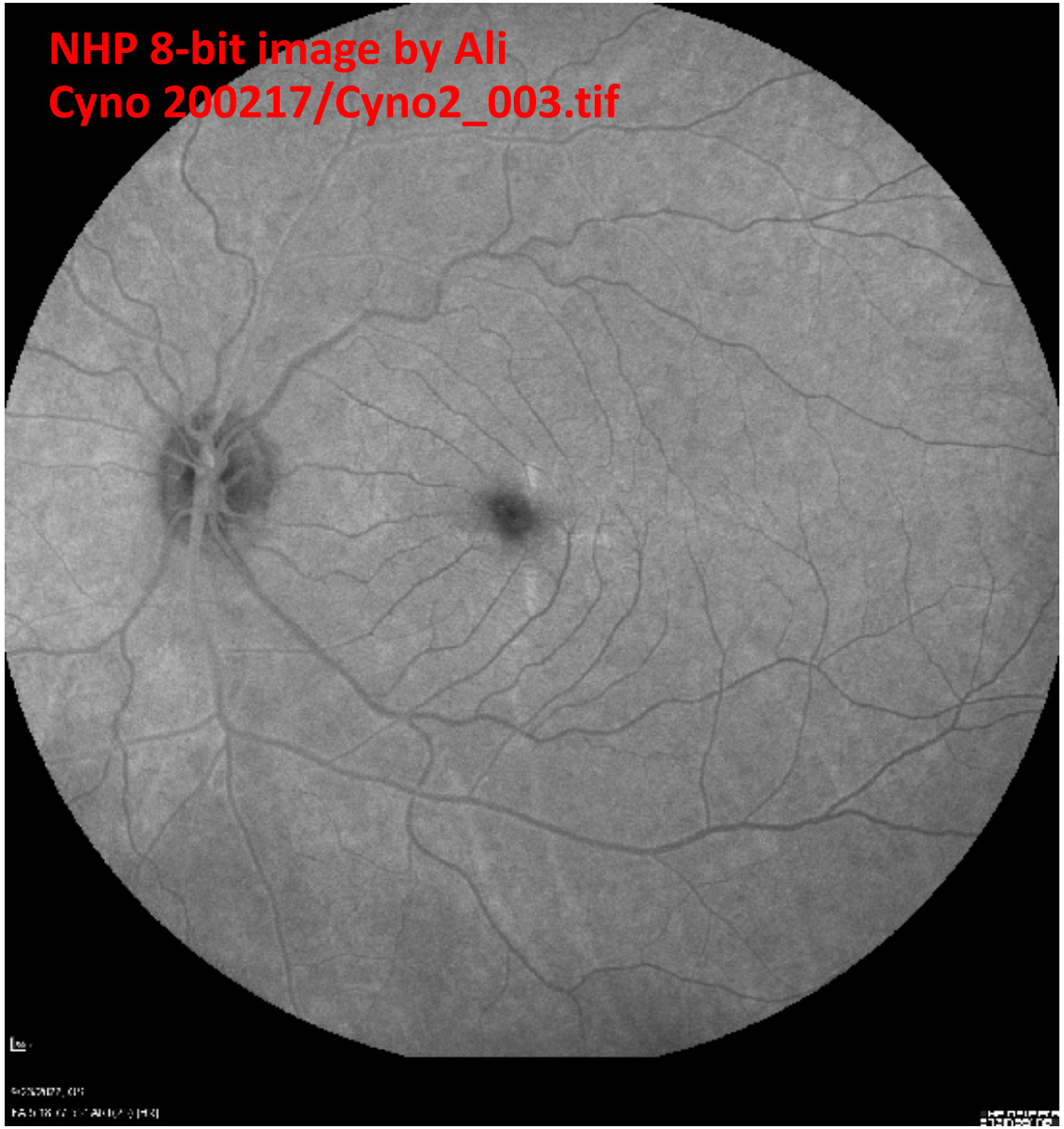


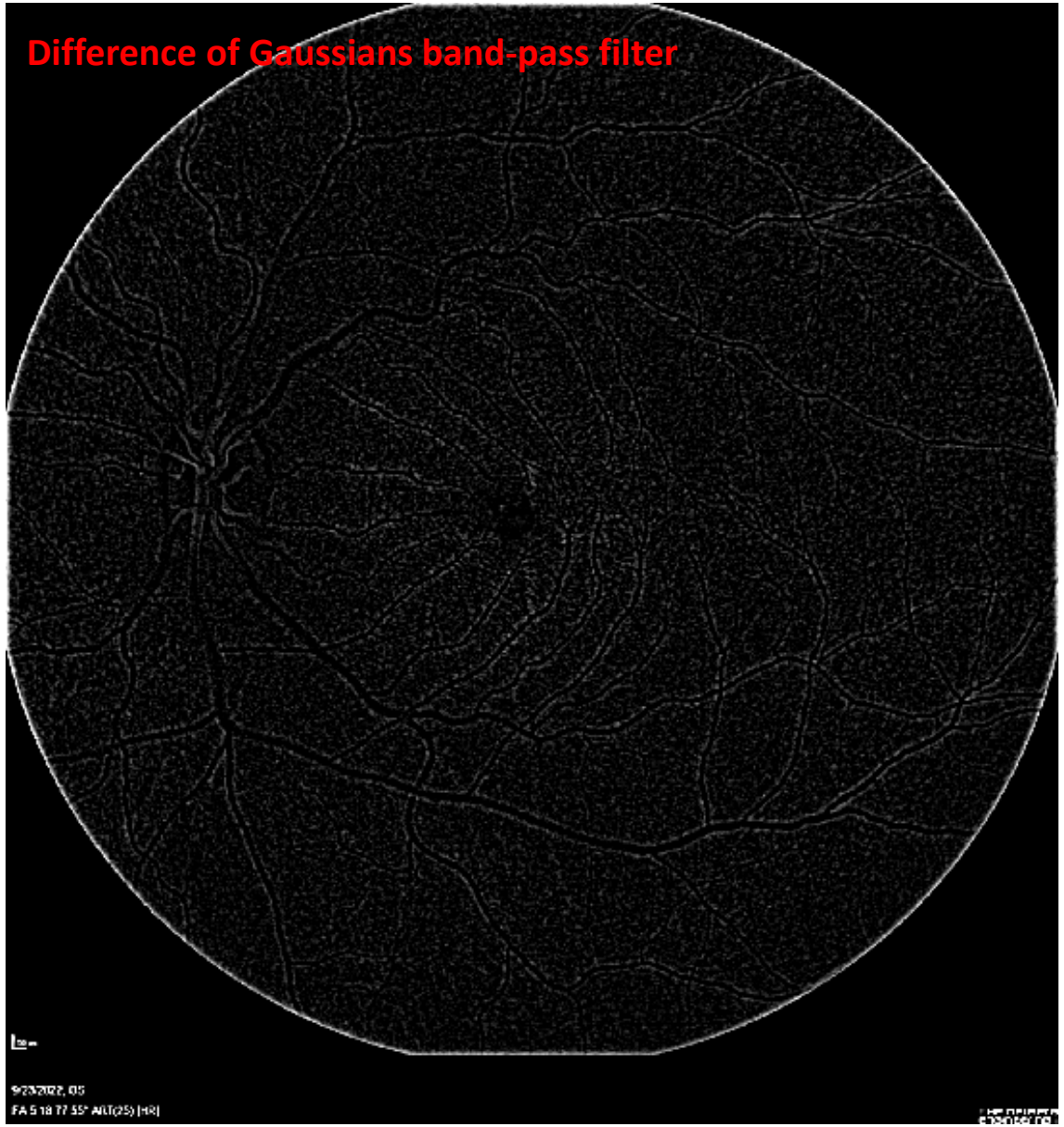
Preliminary analysis



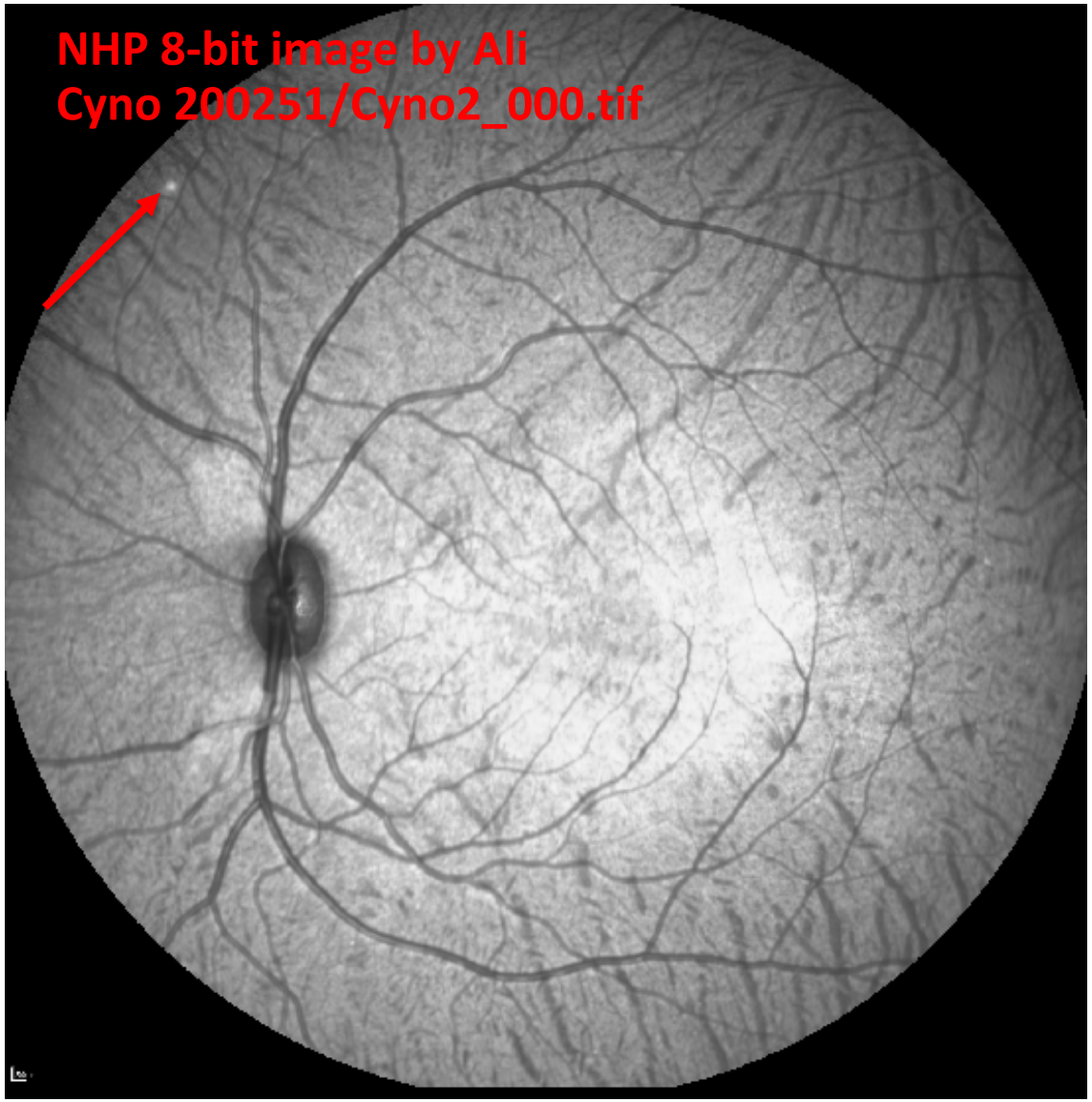


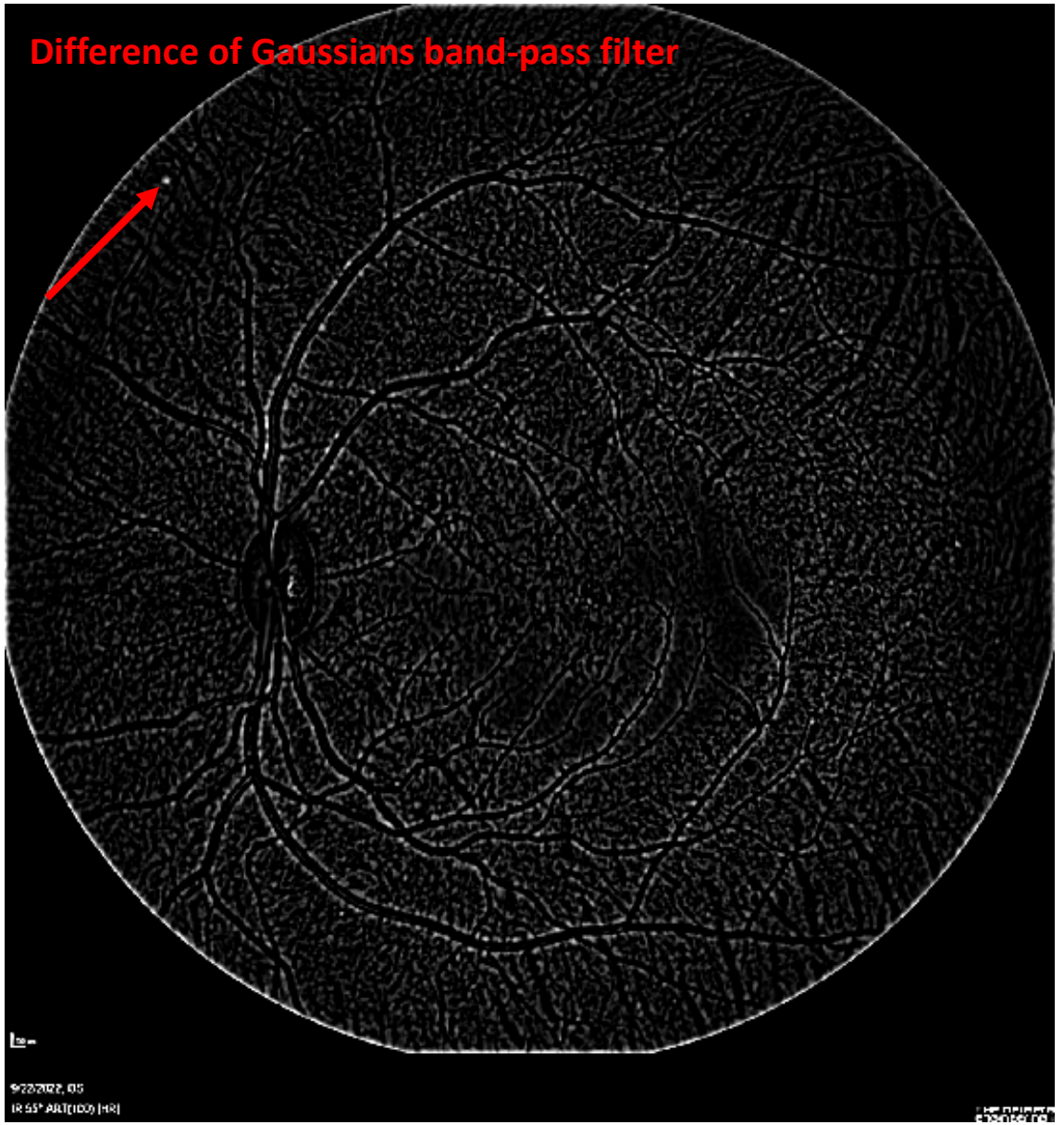
Preliminary analysis





Preliminary analysis





Preliminary analysis

